Usine Frimo 537

Photovoltaic installation and energy storage study

533	383.76 kWp	32	9
Modules	Installed DC power	Strings	Installation zones

Module equipment	**720 W**	Module tilt	**8 degrees**
Site coordinates	**43.55 N / 10.31 E**	Annual shading study	**2026**

Report contents

Engineering report - drawings, schedules and calculation results

Section	Starts on page
01 / OVERVIEW	2
02 / DRAWINGS	4
03 / STRINGS	92
04 / GEOMETRY	97
05 / TECHNICAL INPUTS	98
06 / ANNUAL SHADING	104
07 / CABLING	108
08 / ENERGY AND BESS	111
09 / SPREADSHEET	121
10 / BLOCK DETAILS	122
11 / REVIEW	131

Project summary and data status

Recorded inputs and calculation availability

Parameter	Recorded value	Status
Modules	533	Recorded
Installed DC power	383.760 kWp	Recorded
Module rating	720.0 W	Recorded
Module dimensions	1303.0 x 2384.0 mm	Recorded
Inclination / azimuth	8.0 / 180.0 deg	Recorded
Location	43.55 N / 10.31 E	Recorded
AC active inverter power	125.0 kW	Recorded
Cable linear mass	-	Cable mass unavailable
Report shading study	2026 / Europe/Rome	Reused calculated study

Source roof plan



Source roof plan.

Installation layout

Modules and installation areas

B1: 19.80 x 16.69 m

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105

B2: 19.80 x 16.69 m

106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
196	197	198	199	200	201	202	203	204	205	206	207	208	209	210

B3: 19.80 x 16.69 m

211	212	213	214	215	216	217	218	219	220	221	222	223	224	225
226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
241	242	243	244	245	246	247	248	249	250	251	252	253	254	255
256	257	258	259	260	261	262	263	264	265	266	267	268	269	270
271	272	273	274	275	276	277	278	279	280	281	282	283	284	285
286	287	288	289	290	291	292	293	294	295	296	297	298	299	300
301	302	303	304	305	306	307	308	309	310	311	312	313	314	315

B4: 13.19 x 16.69 m

316	317	318	319	320	321	322	323	324	325
326	327	328	329	330	331	332	333	334	335
336	337	338	339	340	341	342	343	344	345
346	347	348	349	350	351	352	353	354	355
356	357	358	359	360	361	362	363	364	365
366	367	368	369	370	371	372	373	374	375
376	377	378	379	380	381	382	383	384	385

B5: 18.48 x 11.92 m

386	387	388	389	390	391							392	393
394	395	396							397	398	399	400	401
						402	403	404	405	406	407	408	409
		410	411	412	413	414	415	416	417	418	419	420	421
422	423	424	425	426	427	428	429	430	431	432	433	434	435

B6: 18.48 x 4.77 m

	436	437	438	439	440	441	442	443	444	445	446	447	448
449	450	451	452	453	454	455	456	457	458	459	460	461	462

B7: 18.48 x 4.77 m

463	464	465	466	467	468	469	470	471	472	473	474	475	476
477	478	479	480	481	482	483	484	485	486	487	488	489	490

B8: 17.16 x 4.77 m

491	492	493	494	495	496	497	498	499	500	501	502	503
504	505	506	507	508	509	510	511	512	513	514	515	516

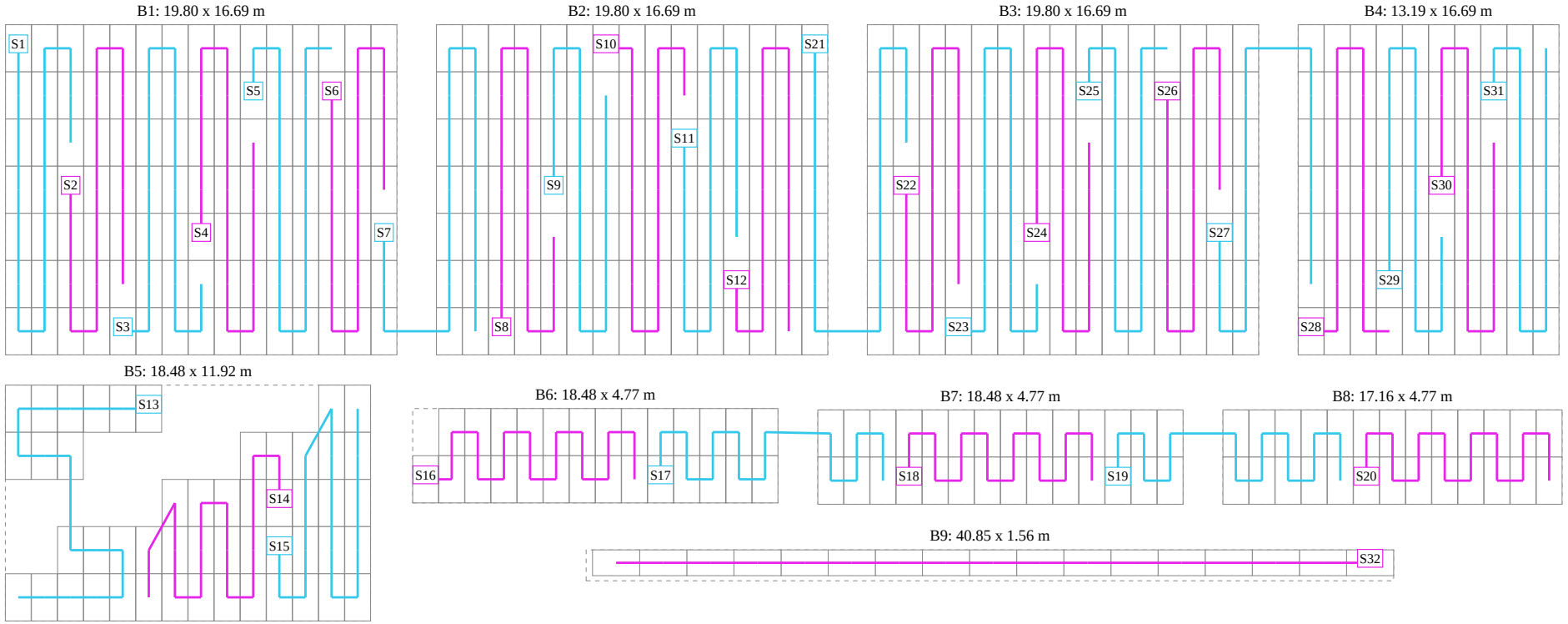
B9: 40.85 x 1.56 m

533	532	531	530	529	528	527	526	525	524	523	522	521	520	519	518	517

Zone dimensions in metres.

String connections

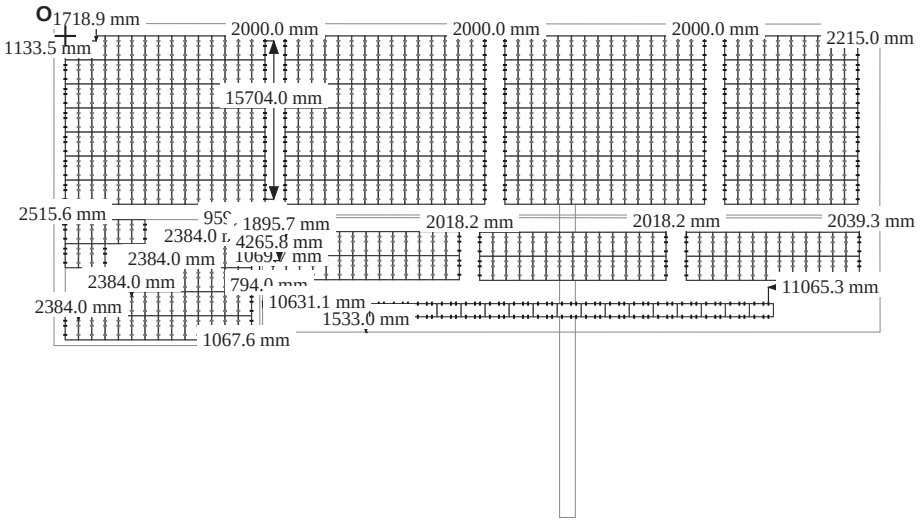
Electrical connection order by string



S1 = String 1, etc. Zone dimensions in metres.

Mounting / dimensioned general plan

Same module IDs as Layout; actual clamp footprints and dimensions in mm



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

10000 mm

MOUNTING / GENERAL PLAN

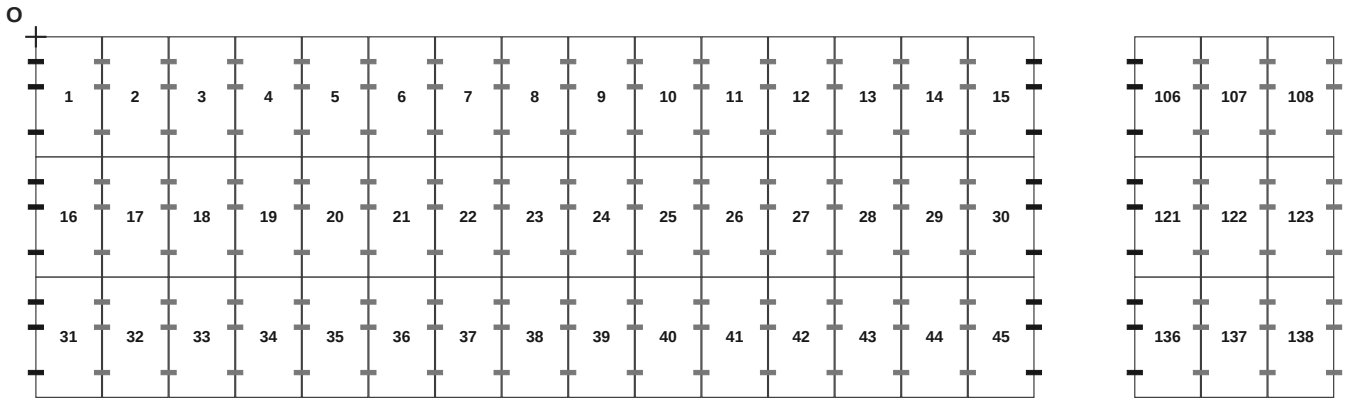
Usine Frimo 537

MT-01 | 29/10/2026 | Ref. 644682F4

1:750 at 100% | dimensions in mm

Datum O: TEMPORARY, corner 1 of module 1. X rightwards and Y downwards. Overall array X/Y dimensions, clear distances on the canvas and clamp positions in the details. Print at 100% to preserve scale.

Mounting / enlarged view 1-1



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

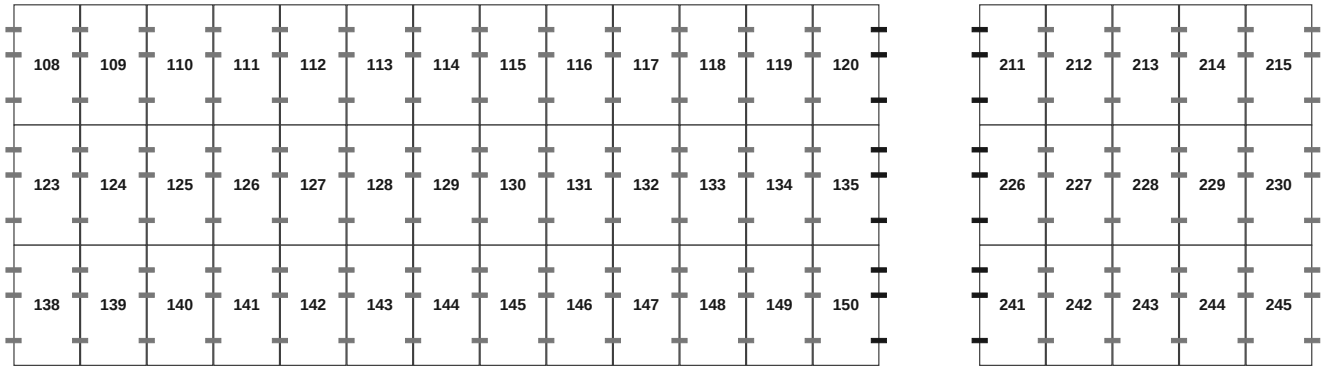
Usine Frimo 537

MT-T01 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 1-2



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

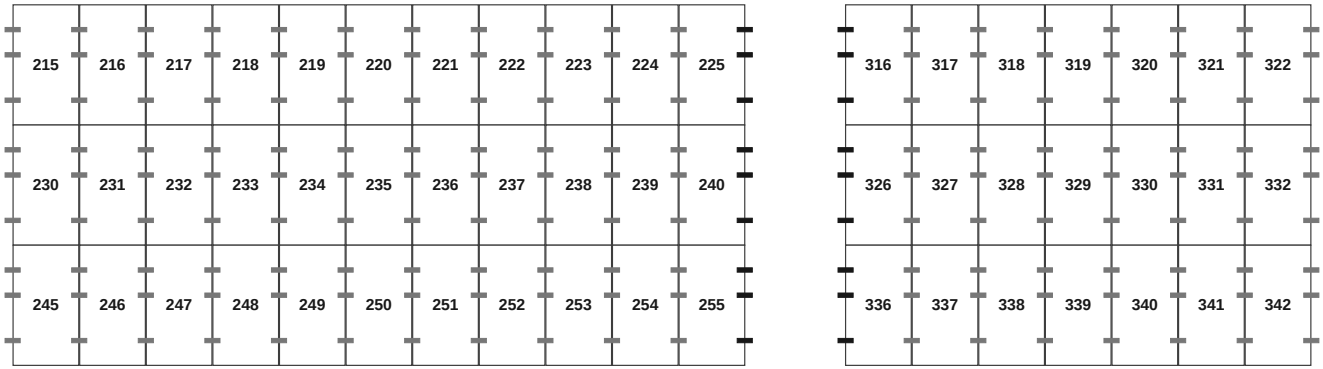
Usine Frimo 537

MT-T02 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 1-3



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

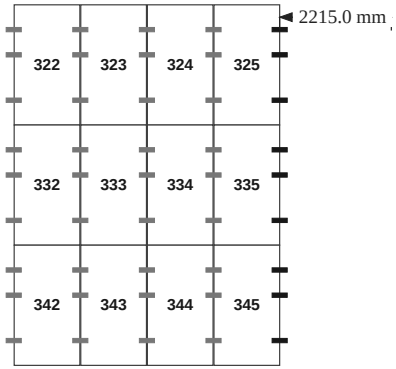
Usine Frimo 537

MT-T03 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 1-4



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

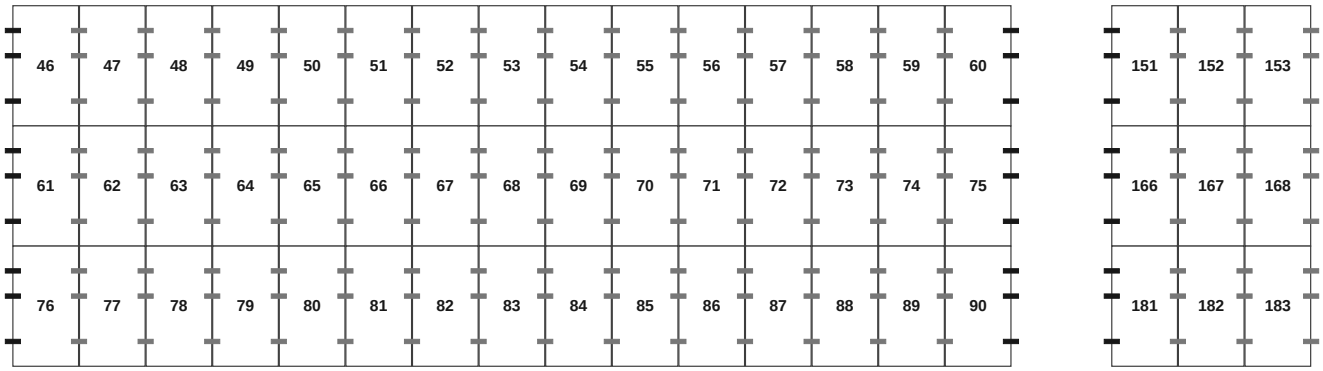
Usine Frimo 537

MT-T04 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 2-1



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

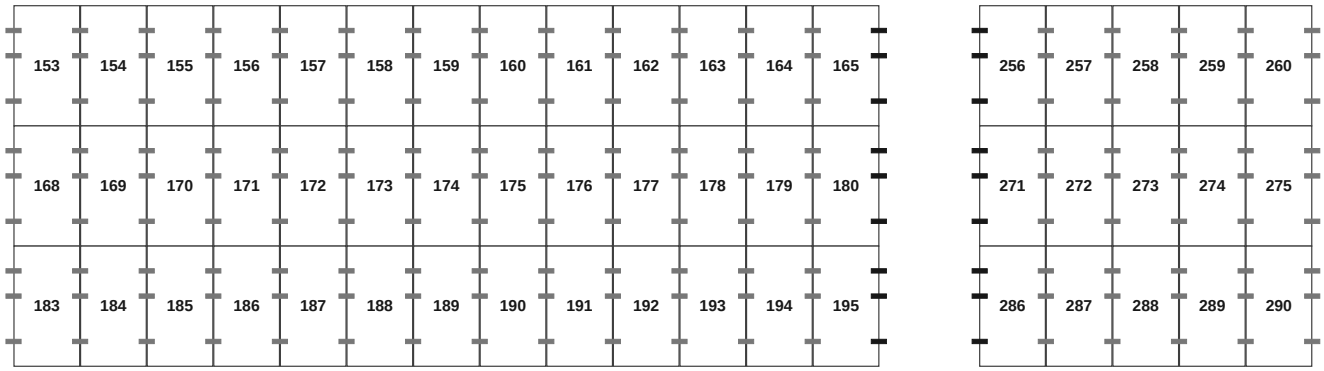
Usine Frimo 537

MT-T05 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 2-2



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

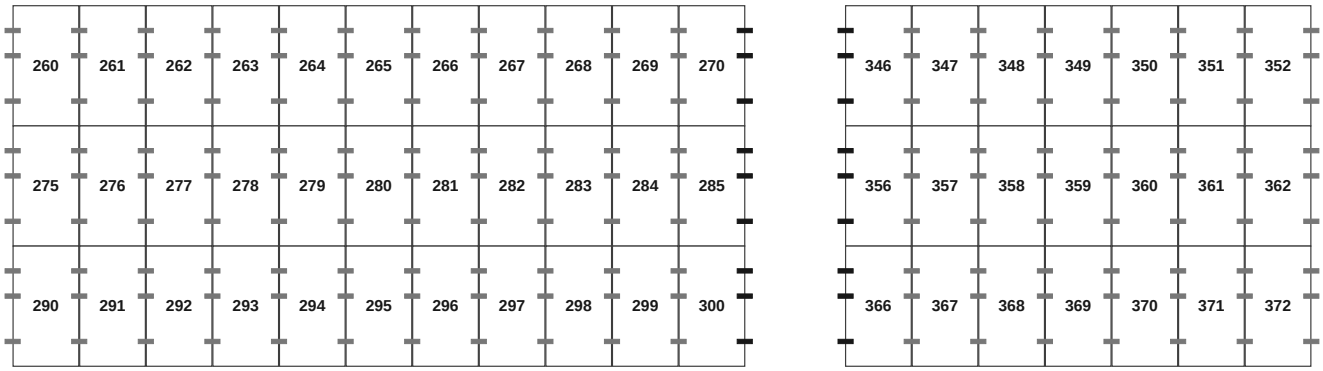
Usine Frimo 537

MT-T06 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 2-3



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

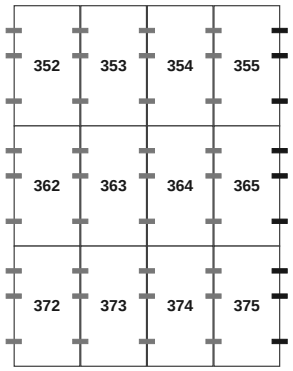
Usine Frimo 537

MT-T07 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 2-4



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

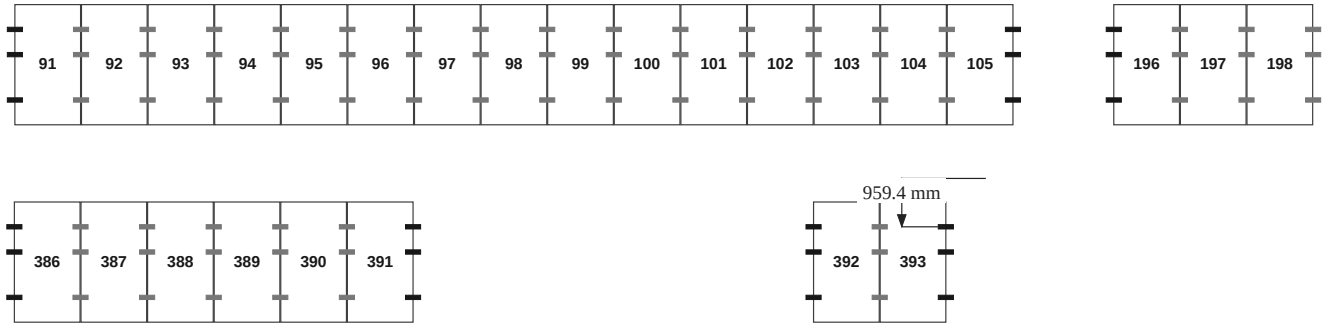
Usine Frimo 537

MT-T08 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 3-1



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

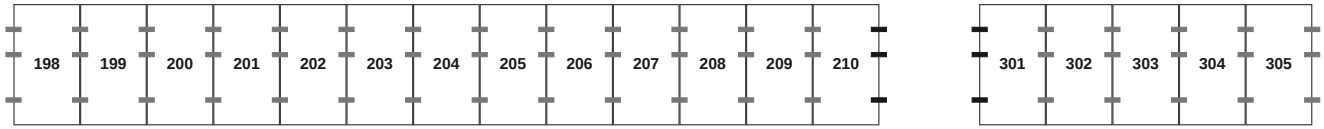
Usine Frimo 537

MT-T09 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 3-2



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

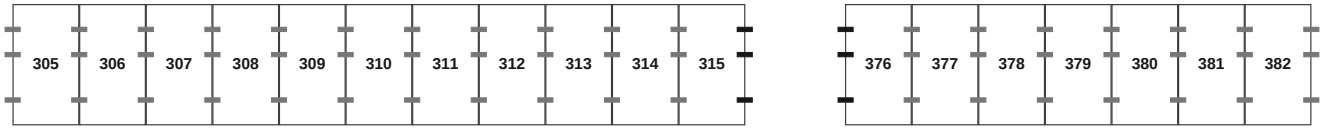
Usine Frimo 537

MT-T10 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 3-3



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

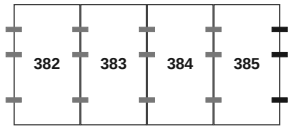
Usine Frimo 537

MT-T11 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 3-4



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

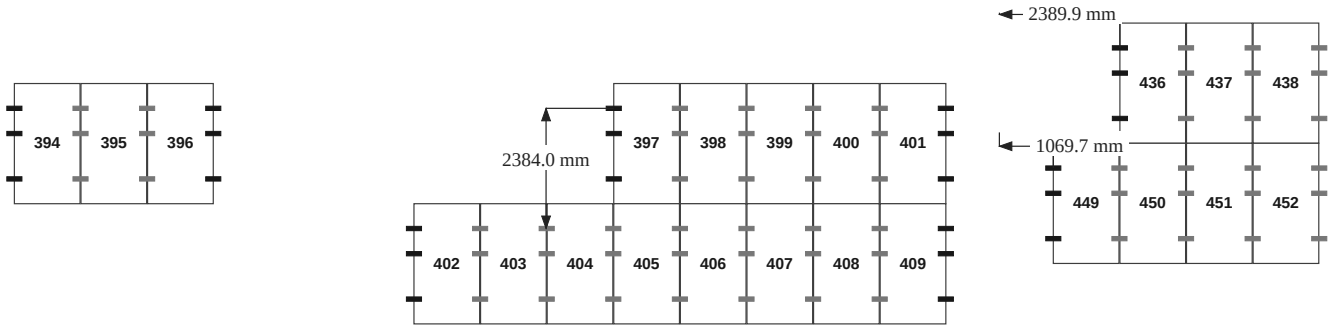
Usine Frimo 537

MT-T12 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 4-1



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

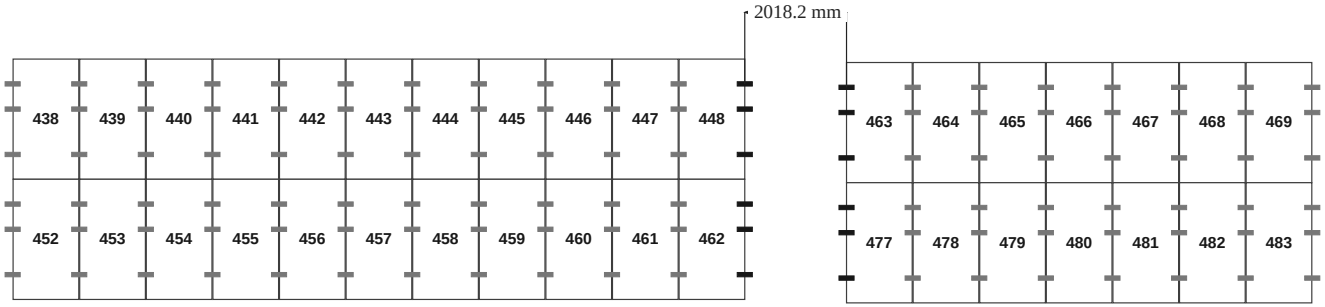
Usine Frimo 537

MT-T13 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 4-2



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

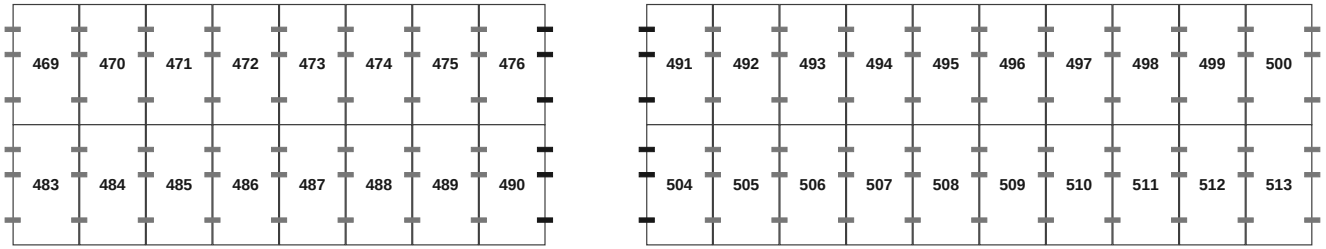
Usine Frimo 537

MT-T14 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 4-3



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

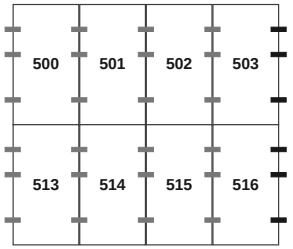
Usine Frimo 537

MT-T15 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 4-4



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

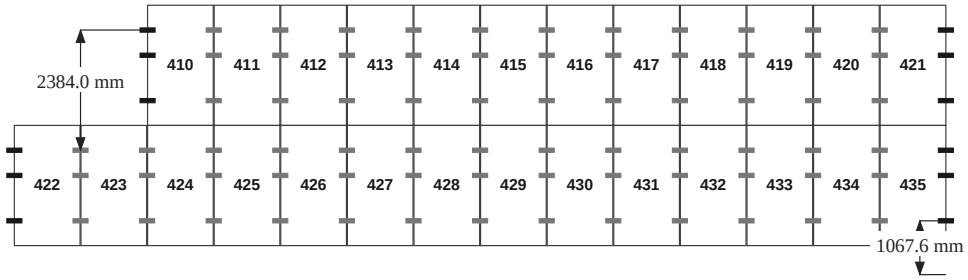
Usine Frimo 537

MT-T16 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 5-1



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

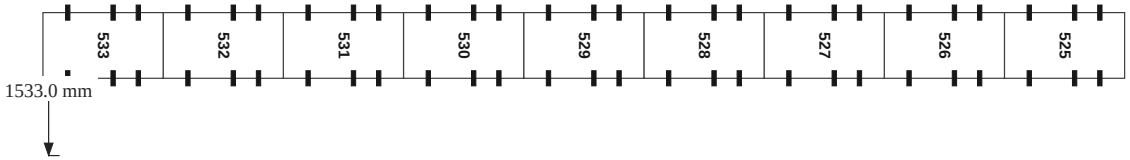
Usine Frimo 537

MT-T17 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 5-2



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

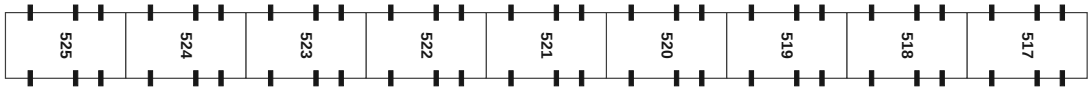
Usine Frimo 537

MT-T18 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / enlarged view 5-3



Shared intermediate clamp: ONE part between TWO frames.

R: end (black) I: intermediate (gray) S: uniform (light gray) !: missing mounting

2000 mm

MOUNTING / ENLARGED VIEW

Usine Frimo 537

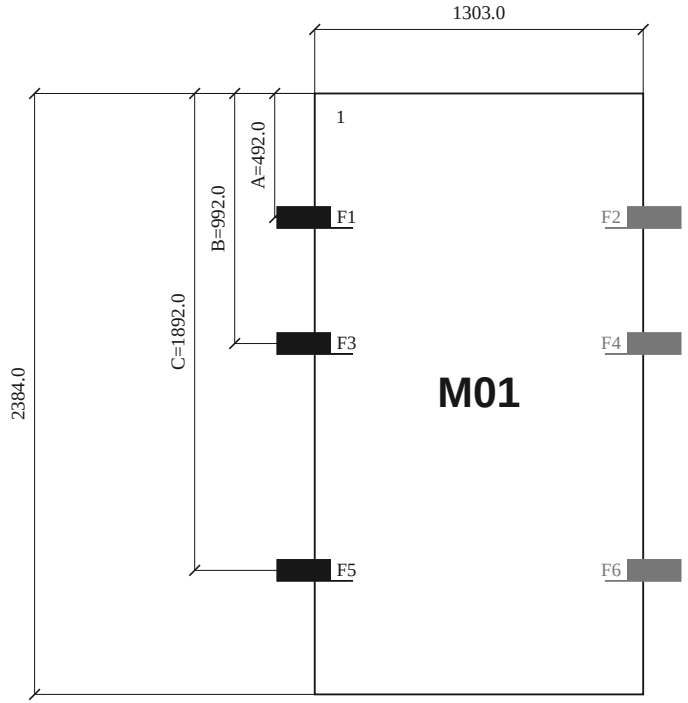
MT-T19 | 29/10/2026 | Ref. 644682F4

1:150 at 100% | dimensions in mm

Adjacent enlarged sheets overlap. Exact coordinates are listed in the placement schedule.

Mounting / detail M01

Local module orientation; corner 1 indicated in the detail



CLAMP CONTACTS / AXES

Point	Side	Axe mm	Type	Lf × lf mm
F1 / A	1	492.0	R	86 × 300
F2 / A	2	492.0	I	86 × 300
F3 / B	1	992.0	R	86 × 300
F4 / B	2	992.0	I	86 × 300
F5 / C	1	1892.0	R	86 × 300
F6 / C	2	1892.0	I	86 × 300

A/B/C: three dimensions from top edge to clamp axes.

Side 1: left / top; side 2: right / bottom.

Lf: parallel to edge; lf: perpendicular to edge.

Shared I contacts: one physical clamp between two modules.

MOUNTING / DETAIL M01

Usine Frimo 537

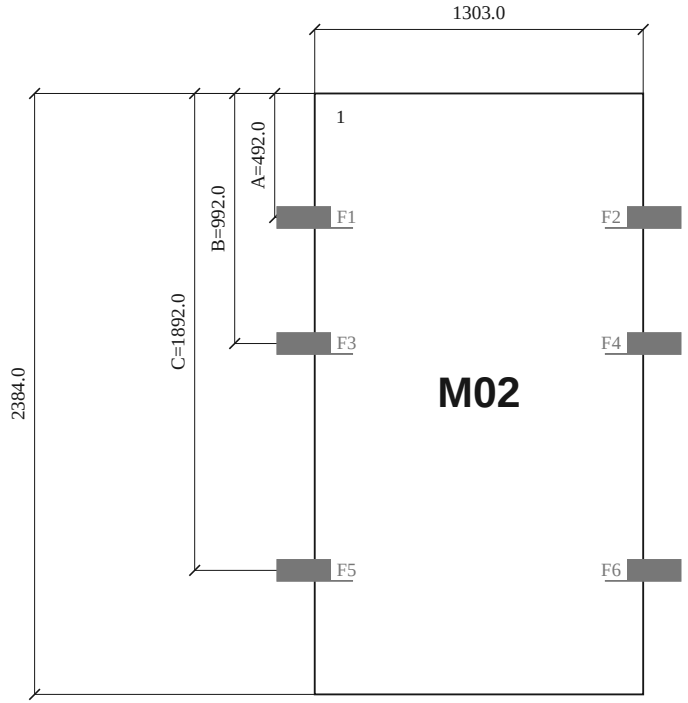
MT-M01 | 29/10/2026 | Ref. 644682F4

1:30 at 100% | dimensions in mm

Modules: 1, 16, 31, 46, 61, 76, 91, 106, 121, 136, 151, 166, 181, 196, 211, 226, 241, 256, 271, 286, 301, 316, 326, 336, 346, 356, 366, 376, 386, 392… (see placement schedule for remaining modules). The clamped edge starts at top for vertical mounting, left for horizontal mounting. With six clamps, A/B/C are three independent distances from the top edge to the clamp centerlines.

Mounting / detail M02

Local module orientation; corner 1 indicated in the detail



CLAMP CONTACTS / AXES

Point	Side	Axe mm	Type	Lf × lf mm
F1 / A	1	492.0	I	86 × 300
F2 / A	2	492.0	I	86 × 300
F3 / B	1	992.0	I	86 × 300
F4 / B	2	992.0	I	86 × 300
F5 / C	1	1892.0	I	86 × 300
F6 / C	2	1892.0	I	86 × 300

A/B/C: three dimensions from top edge to clamp axes.

Side 1: left / top; side 2: right / bottom.

Lf: parallel to edge; lf: perpendicular to edge.

Shared I contacts: one physical clamp between two modules.

MOUNTING / DETAIL M02

Usine Frimo 537

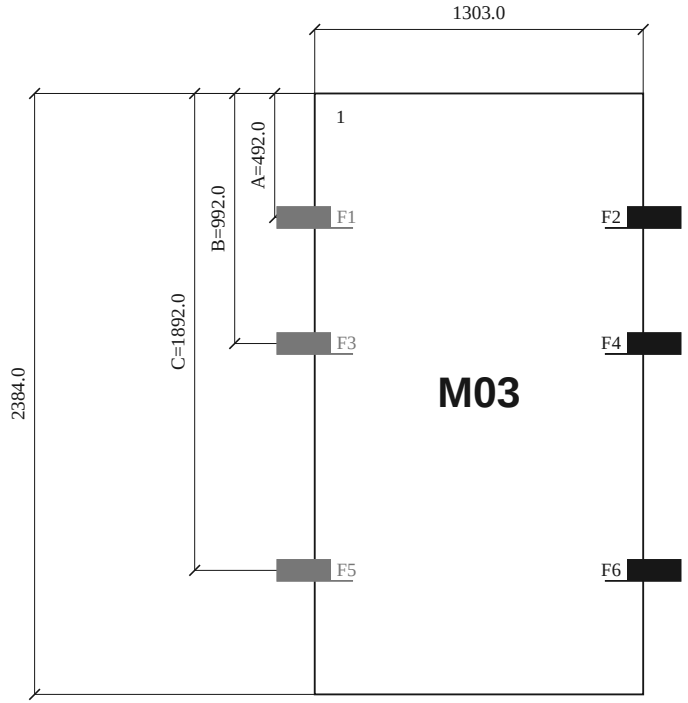
MT-M02 | 29/10/2026 | Ref. 644682F4

1:30 at 100% | dimensions in mm

Modules: 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43… (see placement schedule for remaining modules). The clamped edge starts at top for vertical mounting, left for horizontal mounting. With six clamps, A/B/C are three independent distances from the top edge to the clamp centerlines.

Mounting / detail M03

Local module orientation; corner 1 indicated in the detail



CLAMP CONTACTS / AXES

Point	Side	Axe mm	Type	Lf × lf mm
F1 / A	1	492.0	I	86 × 300
F2 / A	2	492.0	R	86 × 300
F3 / B	1	992.0	I	86 × 300
F4 / B	2	992.0	R	86 × 300
F5 / C	1	1892.0	I	86 × 300
F6 / C	2	1892.0	R	86 × 300

A/B/C: three dimensions from top edge to clamp axes.

Side 1: left / top; side 2: right / bottom.

Lf: parallel to edge; lf: perpendicular to edge.

Shared I contacts: one physical clamp between two modules.

MOUNTING / DETAIL M03

Usine Frimo 537

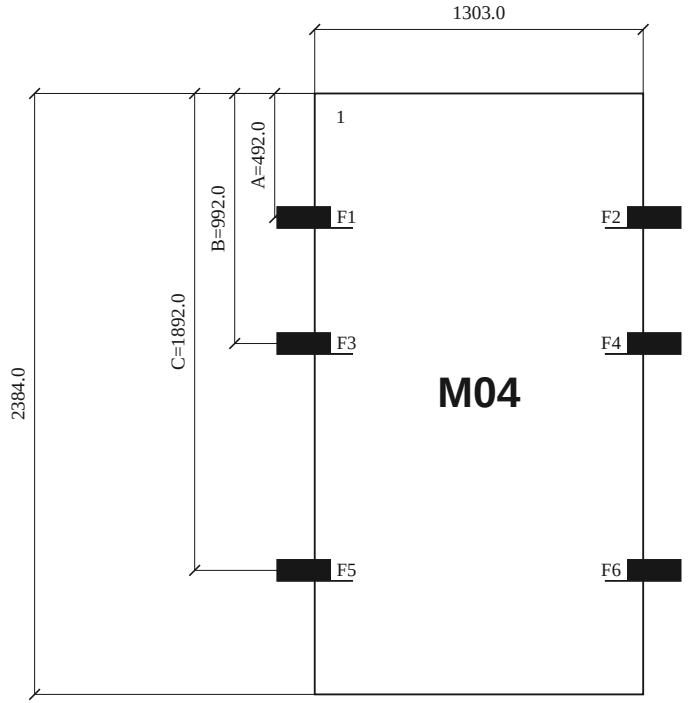
MT-M03 | 29/10/2026 | Ref. 644682F4

1:30 at 100% | dimensions in mm

Modules: 15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165, 180, 195, 210, 225, 240, 255, 270, 285, 300, 315, 325, 335, 345, 355, 365, 375, 385, 391, 393… (see placement schedule for remaining modules). The clamped edge starts at top for vertical mounting, left for horizontal mounting. With six clamps, A/B/C are three independent distances from the top edge to the clamp centerlines.

Mounting / detail M04

Local module orientation; corner 1 indicated in the detail



CLAMP CONTACTS / AXES

Point	Side	Axe mm	Type	Lf × lf mm
F1 / A	1	492.0	R	86 × 300
F2 / A	2	492.0	R	86 × 300
F3 / B	1	992.0	R	86 × 300
F4 / B	2	992.0	R	86 × 300
F5 / C	1	1892.0	R	86 × 300
F6 / C	2	1892.0	R	86 × 300

A/B/C: three dimensions from top edge to clamp axes.

Side 1: left / top; side 2: right / bottom.

Lf: parallel to edge; lf: perpendicular to edge.

Shared I contacts: one physical clamp between two modules.

MOUNTING / DETAIL M04

Usine Frimo 537

MT-M04 | 29/10/2026 | Ref. 644682F4

1:30 at 100% | dimensions in mm

Modules: 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533. The clamped edge starts at top for vertical mounting, left for horizontal mounting. With six clamps, A/B/C are three independent distances from the top edge to the clamp centerlines.

Mounting / clamp references

Type	Reference	Lf mm	lf mm	Contacts	Pieces
R	Rive	86	300	348	348
I	Intermédiaire	86	300	2850	1425

Physical bill of materials: 348 end clamps, 1425 intermediate clamps, 0 uniform clamps, total 1773 pièces. A compatible shared intermediate clamp is counted once. Verify on site.

Drawings show configured clamp footprints and axes. Rails, anchors and drilling positions are not defined by these data.

Mounting / placement 1–18

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
1	M01	0.0	0.0	0.00	6
2	M02	1321.0	0.0	0.00	6
3	M02	2642.0	0.0	0.00	6
4	M02	3963.0	0.0	0.00	6
5	M02	5284.0	0.0	0.00	6
6	M02	6605.0	0.0	0.00	6
7	M02	7926.0	0.0	0.00	6
8	M02	9247.0	0.0	0.00	6
9	M02	10568.0	0.0	0.00	6
10	M02	11889.0	0.0	0.00	6
11	M02	13210.0	0.0	0.00	6
12	M02	14531.0	0.0	0.00	6
13	M02	15852.0	0.0	0.00	6
14	M02	17173.0	0.0	0.00	6
15	M03	18494.0	0.0	0.00	6
16	M01	0.0	2384.0	0.00	6
17	M02	1321.0	2384.0	0.00	6
18	M02	2642.0	2384.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 19–36

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
19	M02	3963.0	2384.0	0.00	6
20	M02	5284.0	2384.0	0.00	6
21	M02	6605.0	2384.0	0.00	6
22	M02	7926.0	2384.0	0.00	6
23	M02	9247.0	2384.0	0.00	6
24	M02	10568.0	2384.0	0.00	6
25	M02	11889.0	2384.0	0.00	6
26	M02	13210.0	2384.0	0.00	6
27	M02	14531.0	2384.0	0.00	6
28	M02	15852.0	2384.0	0.00	6
29	M02	17173.0	2384.0	0.00	6
30	M03	18494.0	2384.0	0.00	6
31	M01	0.0	4768.0	0.00	6
32	M02	1321.0	4768.0	0.00	6
33	M02	2642.0	4768.0	0.00	6
34	M02	3963.0	4768.0	0.00	6
35	M02	5284.0	4768.0	0.00	6
36	M02	6605.0	4768.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 37–54

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
37	M02	7926.0	4768.0	0.00	6
38	M02	9247.0	4768.0	0.00	6
39	M02	10568.0	4768.0	0.00	6
40	M02	11889.0	4768.0	0.00	6
41	M02	13210.0	4768.0	0.00	6
42	M02	14531.0	4768.0	0.00	6
43	M02	15852.0	4768.0	0.00	6
44	M02	17173.0	4768.0	0.00	6
45	M03	18494.0	4768.0	0.00	6
46	M01	0.0	7152.0	0.00	6
47	M02	1321.0	7152.0	0.00	6
48	M02	2642.0	7152.0	0.00	6
49	M02	3963.0	7152.0	0.00	6
50	M02	5284.0	7152.0	0.00	6
51	M02	6605.0	7152.0	0.00	6
52	M02	7926.0	7152.0	0.00	6
53	M02	9247.0	7152.0	0.00	6
54	M02	10568.0	7152.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 55–72

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
55	M02	11889.0	7152.0	0.00	6
56	M02	13210.0	7152.0	0.00	6
57	M02	14531.0	7152.0	0.00	6
58	M02	15852.0	7152.0	0.00	6
59	M02	17173.0	7152.0	0.00	6
60	M03	18494.0	7152.0	0.00	6
61	M01	0.0	9536.0	0.00	6
62	M02	1321.0	9536.0	0.00	6
63	M02	2642.0	9536.0	0.00	6
64	M02	3963.0	9536.0	0.00	6
65	M02	5284.0	9536.0	0.00	6
66	M02	6605.0	9536.0	0.00	6
67	M02	7926.0	9536.0	0.00	6
68	M02	9247.0	9536.0	0.00	6
69	M02	10568.0	9536.0	0.00	6
70	M02	11889.0	9536.0	0.00	6
71	M02	13210.0	9536.0	0.00	6
72	M02	14531.0	9536.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 73–90

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
73	M02	15852.0	9536.0	0.00	6
74	M02	17173.0	9536.0	0.00	6
75	M03	18494.0	9536.0	0.00	6
76	M01	0.0	11920.0	0.00	6
77	M02	1321.0	11920.0	0.00	6
78	M02	2642.0	11920.0	0.00	6
79	M02	3963.0	11920.0	0.00	6
80	M02	5284.0	11920.0	0.00	6
81	M02	6605.0	11920.0	0.00	6
82	M02	7926.0	11920.0	0.00	6
83	M02	9247.0	11920.0	0.00	6
84	M02	10568.0	11920.0	0.00	6
85	M02	11889.0	11920.0	0.00	6
86	M02	13210.0	11920.0	0.00	6
87	M02	14531.0	11920.0	0.00	6
88	M02	15852.0	11920.0	0.00	6
89	M02	17173.0	11920.0	0.00	6
90	M03	18494.0	11920.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 91–108

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
91	M01	0.0	14304.0	0.00	6
92	M02	1321.0	14304.0	0.00	6
93	M02	2642.0	14304.0	0.00	6
94	M02	3963.0	14304.0	0.00	6
95	M02	5284.0	14304.0	0.00	6
96	M02	6605.0	14304.0	0.00	6
97	M02	7926.0	14304.0	0.00	6
98	M02	9247.0	14304.0	0.00	6
99	M02	10568.0	14304.0	0.00	6
100	M02	11889.0	14304.0	0.00	6
101	M02	13210.0	14304.0	0.00	6
102	M02	14531.0	14304.0	0.00	6
103	M02	15852.0	14304.0	0.00	6
104	M02	17173.0	14304.0	0.00	6
105	M03	18494.0	14304.0	0.00	6
106	M01	21797.0	-0.0	0.00	6
107	M02	23118.0	-0.0	0.00	6
108	M02	24439.0	-0.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 109–126

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
109	M02	25760.0	-0.0	0.00	6
110	M02	27081.0	-0.0	0.00	6
111	M02	28402.0	-0.0	0.00	6
112	M02	29723.0	-0.0	0.00	6
113	M02	31044.0	-0.0	0.00	6
114	M02	32365.0	-0.0	0.00	6
115	M02	33686.0	-0.0	0.00	6
116	M02	35007.0	-0.0	0.00	6
117	M02	36328.0	-0.0	0.00	6
118	M02	37649.0	-0.0	0.00	6
119	M02	38970.0	-0.0	0.00	6
120	M03	40291.0	-0.0	0.00	6
121	M01	21797.0	2384.0	0.00	6
122	M02	23118.0	2384.0	0.00	6
123	M02	24439.0	2384.0	0.00	6
124	M02	25760.0	2384.0	0.00	6
125	M02	27081.0	2384.0	0.00	6
126	M02	28402.0	2384.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 127–144

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
127	M02	29723.0	2384.0	0.00	6
128	M02	31044.0	2384.0	0.00	6
129	M02	32365.0	2384.0	0.00	6
130	M02	33686.0	2384.0	0.00	6
131	M02	35007.0	2384.0	0.00	6
132	M02	36328.0	2384.0	0.00	6
133	M02	37649.0	2384.0	0.00	6
134	M02	38970.0	2384.0	0.00	6
135	M03	40291.0	2384.0	0.00	6
136	M01	21797.0	4768.0	0.00	6
137	M02	23118.0	4768.0	0.00	6
138	M02	24439.0	4768.0	0.00	6
139	M02	25760.0	4768.0	0.00	6
140	M02	27081.0	4768.0	0.00	6
141	M02	28402.0	4768.0	0.00	6
142	M02	29723.0	4768.0	0.00	6
143	M02	31044.0	4768.0	0.00	6
144	M02	32365.0	4768.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 145–162

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
145	M02	33686.0	4768.0	0.00	6
146	M02	35007.0	4768.0	0.00	6
147	M02	36328.0	4768.0	0.00	6
148	M02	37649.0	4768.0	0.00	6
149	M02	38970.0	4768.0	0.00	6
150	M03	40291.0	4768.0	0.00	6
151	M01	21797.0	7152.0	0.00	6
152	M02	23118.0	7152.0	0.00	6
153	M02	24439.0	7152.0	0.00	6
154	M02	25760.0	7152.0	0.00	6
155	M02	27081.0	7152.0	0.00	6
156	M02	28402.0	7152.0	0.00	6
157	M02	29723.0	7152.0	0.00	6
158	M02	31044.0	7152.0	0.00	6
159	M02	32365.0	7152.0	0.00	6
160	M02	33686.0	7152.0	0.00	6
161	M02	35007.0	7152.0	0.00	6
162	M02	36328.0	7152.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 163–180

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
163	M02	37649.0	7152.0	0.00	6
164	M02	38970.0	7152.0	0.00	6
165	M03	40291.0	7152.0	0.00	6
166	M01	21797.0	9536.0	0.00	6
167	M02	23118.0	9536.0	0.00	6
168	M02	24439.0	9536.0	0.00	6
169	M02	25760.0	9536.0	0.00	6
170	M02	27081.0	9536.0	0.00	6
171	M02	28402.0	9536.0	0.00	6
172	M02	29723.0	9536.0	0.00	6
173	M02	31044.0	9536.0	0.00	6
174	M02	32365.0	9536.0	0.00	6
175	M02	33686.0	9536.0	0.00	6
176	M02	35007.0	9536.0	0.00	6
177	M02	36328.0	9536.0	0.00	6
178	M02	37649.0	9536.0	0.00	6
179	M02	38970.0	9536.0	0.00	6
180	M03	40291.0	9536.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 181–198

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
181	M01	21797.0	11920.0	0.00	6
182	M02	23118.0	11920.0	0.00	6
183	M02	24439.0	11920.0	0.00	6
184	M02	25760.0	11920.0	0.00	6
185	M02	27081.0	11920.0	0.00	6
186	M02	28402.0	11920.0	0.00	6
187	M02	29723.0	11920.0	0.00	6
188	M02	31044.0	11920.0	0.00	6
189	M02	32365.0	11920.0	0.00	6
190	M02	33686.0	11920.0	0.00	6
191	M02	35007.0	11920.0	0.00	6
192	M02	36328.0	11920.0	0.00	6
193	M02	37649.0	11920.0	0.00	6
194	M02	38970.0	11920.0	0.00	6
195	M03	40291.0	11920.0	0.00	6
196	M01	21797.0	14304.0	0.00	6
197	M02	23118.0	14304.0	0.00	6
198	M02	24439.0	14304.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 199–216

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
199	M02	25760.0	14304.0	0.00	6
200	M02	27081.0	14304.0	0.00	6
201	M02	28402.0	14304.0	0.00	6
202	M02	29723.0	14304.0	0.00	6
203	M02	31044.0	14304.0	0.00	6
204	M02	32365.0	14304.0	0.00	6
205	M02	33686.0	14304.0	0.00	6
206	M02	35007.0	14304.0	0.00	6
207	M02	36328.0	14304.0	0.00	6
208	M02	37649.0	14304.0	0.00	6
209	M02	38970.0	14304.0	0.00	6
210	M03	40291.0	14304.0	0.00	6
211	M01	43594.0	-0.0	0.00	6
212	M02	44915.0	-0.0	0.00	6
213	M02	46236.0	-0.0	0.00	6
214	M02	47557.0	-0.0	0.00	6
215	M02	48878.0	-0.0	0.00	6
216	M02	50199.0	-0.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 217–234

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
217	M02	51520.0	-0.0	0.00	6
218	M02	52841.0	-0.0	0.00	6
219	M02	54162.0	-0.0	0.00	6
220	M02	55483.0	-0.0	0.00	6
221	M02	56804.0	-0.0	0.00	6
222	M02	58125.0	-0.0	0.00	6
223	M02	59446.0	-0.0	0.00	6
224	M02	60767.0	-0.0	0.00	6
225	M03	62088.0	-0.0	0.00	6
226	M01	43594.0	2384.0	0.00	6
227	M02	44915.0	2384.0	0.00	6
228	M02	46236.0	2384.0	0.00	6
229	M02	47557.0	2384.0	0.00	6
230	M02	48878.0	2384.0	0.00	6
231	M02	50199.0	2384.0	0.00	6
232	M02	51520.0	2384.0	0.00	6
233	M02	52841.0	2384.0	0.00	6
234	M02	54162.0	2384.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 235–252

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
235	M02	55483.0	2384.0	0.00	6
236	M02	56804.0	2384.0	0.00	6
237	M02	58125.0	2384.0	0.00	6
238	M02	59446.0	2384.0	0.00	6
239	M02	60767.0	2384.0	0.00	6
240	M03	62088.0	2384.0	0.00	6
241	M01	43594.0	4768.0	0.00	6
242	M02	44915.0	4768.0	0.00	6
243	M02	46236.0	4768.0	0.00	6
244	M02	47557.0	4768.0	0.00	6
245	M02	48878.0	4768.0	0.00	6
246	M02	50199.0	4768.0	0.00	6
247	M02	51520.0	4768.0	0.00	6
248	M02	52841.0	4768.0	0.00	6
249	M02	54162.0	4768.0	0.00	6
250	M02	55483.0	4768.0	0.00	6
251	M02	56804.0	4768.0	0.00	6
252	M02	58125.0	4768.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 253–270

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
253	M02	59446.0	4768.0	0.00	6
254	M02	60767.0	4768.0	0.00	6
255	M03	62088.0	4768.0	0.00	6
256	M01	43594.0	7152.0	0.00	6
257	M02	44915.0	7152.0	0.00	6
258	M02	46236.0	7152.0	0.00	6
259	M02	47557.0	7152.0	0.00	6
260	M02	48878.0	7152.0	0.00	6
261	M02	50199.0	7152.0	0.00	6
262	M02	51520.0	7152.0	0.00	6
263	M02	52841.0	7152.0	0.00	6
264	M02	54162.0	7152.0	0.00	6
265	M02	55483.0	7152.0	0.00	6
266	M02	56804.0	7152.0	0.00	6
267	M02	58125.0	7152.0	0.00	6
268	M02	59446.0	7152.0	0.00	6
269	M02	60767.0	7152.0	0.00	6
270	M03	62088.0	7152.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 271–288

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
271	M01	43594.0	9536.0	0.00	6
272	M02	44915.0	9536.0	0.00	6
273	M02	46236.0	9536.0	0.00	6
274	M02	47557.0	9536.0	0.00	6
275	M02	48878.0	9536.0	0.00	6
276	M02	50199.0	9536.0	0.00	6
277	M02	51520.0	9536.0	0.00	6
278	M02	52841.0	9536.0	0.00	6
279	M02	54162.0	9536.0	0.00	6
280	M02	55483.0	9536.0	0.00	6
281	M02	56804.0	9536.0	0.00	6
282	M02	58125.0	9536.0	0.00	6
283	M02	59446.0	9536.0	0.00	6
284	M02	60767.0	9536.0	0.00	6
285	M03	62088.0	9536.0	0.00	6
286	M01	43594.0	11920.0	0.00	6
287	M02	44915.0	11920.0	0.00	6
288	M02	46236.0	11920.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 289–306

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
289	M02	47557.0	11920.0	0.00	6
290	M02	48878.0	11920.0	0.00	6
291	M02	50199.0	11920.0	0.00	6
292	M02	51520.0	11920.0	0.00	6
293	M02	52841.0	11920.0	0.00	6
294	M02	54162.0	11920.0	0.00	6
295	M02	55483.0	11920.0	0.00	6
296	M02	56804.0	11920.0	0.00	6
297	M02	58125.0	11920.0	0.00	6
298	M02	59446.0	11920.0	0.00	6
299	M02	60767.0	11920.0	0.00	6
300	M03	62088.0	11920.0	0.00	6
301	M01	43594.0	14304.0	0.00	6
302	M02	44915.0	14304.0	0.00	6
303	M02	46236.0	14304.0	0.00	6
304	M02	47557.0	14304.0	0.00	6
305	M02	48878.0	14304.0	0.00	6
306	M02	50199.0	14304.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 307–324

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
307	M02	51520.0	14304.0	0.00	6
308	M02	52841.0	14304.0	0.00	6
309	M02	54162.0	14304.0	0.00	6
310	M02	55483.0	14304.0	0.00	6
311	M02	56804.0	14304.0	0.00	6
312	M02	58125.0	14304.0	0.00	6
313	M02	59446.0	14304.0	0.00	6
314	M02	60767.0	14304.0	0.00	6
315	M03	62088.0	14304.0	0.00	6
316	M01	65391.0	-0.0	0.00	6
317	M02	66712.0	-0.0	0.00	6
318	M02	68033.0	-0.0	0.00	6
319	M02	69354.0	-0.0	0.00	6
320	M02	70675.0	-0.0	0.00	6
321	M02	71996.0	-0.0	0.00	6
322	M02	73317.0	-0.0	0.00	6
323	M02	74638.0	-0.0	0.00	6
324	M02	75959.0	-0.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 325–342

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
325	M03	77280.0	-0.0	0.00	6
326	M01	65391.0	2384.0	0.00	6
327	M02	66712.0	2384.0	0.00	6
328	M02	68033.0	2384.0	0.00	6
329	M02	69354.0	2384.0	0.00	6
330	M02	70675.0	2384.0	0.00	6
331	M02	71996.0	2384.0	0.00	6
332	M02	73317.0	2384.0	0.00	6
333	M02	74638.0	2384.0	0.00	6
334	M02	75959.0	2384.0	0.00	6
335	M03	77280.0	2384.0	0.00	6
336	M01	65391.0	4768.0	0.00	6
337	M02	66712.0	4768.0	0.00	6
338	M02	68033.0	4768.0	0.00	6
339	M02	69354.0	4768.0	0.00	6
340	M02	70675.0	4768.0	0.00	6
341	M02	71996.0	4768.0	0.00	6
342	M02	73317.0	4768.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 343–360

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
343	M02	74638.0	4768.0	0.00	6
344	M02	75959.0	4768.0	0.00	6
345	M03	77280.0	4768.0	0.00	6
346	M01	65391.0	7152.0	0.00	6
347	M02	66712.0	7152.0	0.00	6
348	M02	68033.0	7152.0	0.00	6
349	M02	69354.0	7152.0	0.00	6
350	M02	70675.0	7152.0	0.00	6
351	M02	71996.0	7152.0	0.00	6
352	M02	73317.0	7152.0	0.00	6
353	M02	74638.0	7152.0	0.00	6
354	M02	75959.0	7152.0	0.00	6
355	M03	77280.0	7152.0	0.00	6
356	M01	65391.0	9536.0	0.00	6
357	M02	66712.0	9536.0	0.00	6
358	M02	68033.0	9536.0	0.00	6
359	M02	69354.0	9536.0	0.00	6
360	M02	70675.0	9536.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 361–378

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
361	M02	71996.0	9536.0	0.00	6
362	M02	73317.0	9536.0	0.00	6
363	M02	74638.0	9536.0	0.00	6
364	M02	75959.0	9536.0	0.00	6
365	M03	77280.0	9536.0	0.00	6
366	M01	65391.0	11920.0	0.00	6
367	M02	66712.0	11920.0	0.00	6
368	M02	68033.0	11920.0	0.00	6
369	M02	69354.0	11920.0	0.00	6
370	M02	70675.0	11920.0	0.00	6
371	M02	71996.0	11920.0	0.00	6
372	M02	73317.0	11920.0	0.00	6
373	M02	74638.0	11920.0	0.00	6
374	M02	75959.0	11920.0	0.00	6
375	M03	77280.0	11920.0	0.00	6
376	M01	65391.0	14304.0	0.00	6
377	M02	66712.0	14304.0	0.00	6
378	M02	68033.0	14304.0	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 379–396

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
379	M02	69354.0	14304.0	0.00	6
380	M02	70675.0	14304.0	0.00	6
381	M02	71996.0	14304.0	0.00	6
382	M02	73317.0	14304.0	0.00	6
383	M02	74638.0	14304.0	0.00	6
384	M02	75959.0	14304.0	0.00	6
385	M03	77280.0	14304.0	0.00	6
386	M01	-10.1	18219.6	0.00	6
387	M02	1310.9	18219.6	0.00	6
388	M02	2631.9	18219.6	0.00	6
389	M02	3952.9	18219.6	0.00	6
390	M02	5273.9	18219.6	0.00	6
391	M03	6594.9	18219.6	0.00	6
392	M01	15841.9	18219.6	0.00	6
393	M03	17162.9	18219.6	0.00	6
394	M01	-10.1	20603.6	0.00	6
395	M02	1310.9	20603.6	0.00	6
396	M03	2631.9	20603.6	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 397–414

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
397	M01	11878.9	20603.6	0.00	6
398	M02	13199.9	20603.6	0.00	6
399	M02	14520.9	20603.6	0.00	6
400	M02	15841.9	20603.6	0.00	6
401	M03	17162.9	20603.6	0.00	6
402	M01	7915.9	22987.6	0.00	6
403	M02	9236.9	22987.6	0.00	6
404	M02	10557.9	22987.6	0.00	6
405	M02	11878.9	22987.6	0.00	6
406	M02	13199.9	22987.6	0.00	6
407	M02	14520.9	22987.6	0.00	6
408	M02	15841.9	22987.6	0.00	6
409	M03	17162.9	22987.6	0.00	6
410	M01	2631.9	25371.6	0.00	6
411	M02	3952.9	25371.6	0.00	6
412	M02	5273.9	25371.6	0.00	6
413	M02	6594.9	25371.6	0.00	6
414	M02	7915.9	25371.6	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 415–432

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
415	M02	9236.9	25371.6	0.00	6
416	M02	10557.9	25371.6	0.00	6
417	M02	11878.9	25371.6	0.00	6
418	M02	13199.9	25371.6	0.00	6
419	M02	14520.9	25371.6	0.00	6
420	M02	15841.9	25371.6	0.00	6
421	M03	17162.9	25371.6	0.00	6
422	M01	-10.1	27755.6	0.00	6
423	M02	1310.9	27755.6	0.00	6
424	M02	2631.9	27755.6	0.00	6
425	M02	3952.9	27755.6	0.00	6
426	M02	5273.9	27755.6	0.00	6
427	M02	6594.9	27755.6	0.00	6
428	M02	7915.9	27755.6	0.00	6
429	M02	9236.9	27755.6	0.00	6
430	M02	10557.9	27755.6	0.00	6
431	M02	11878.9	27755.6	0.00	6
432	M02	13199.9	27755.6	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 433–450

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
433	M02	14520.9	27755.6	0.00	6
434	M02	15841.9	27755.6	0.00	6
435	M03	17162.9	27755.6	0.00	6
436	M01	21918.3	19404.6	0.00	6
437	M02	23239.3	19404.6	0.00	6
438	M02	24560.3	19404.6	0.00	6
439	M02	25881.3	19404.6	0.00	6
440	M02	27202.3	19404.6	0.00	6
441	M02	28523.3	19404.6	0.00	6
442	M02	29844.3	19404.6	0.00	6
443	M02	31165.3	19404.6	0.00	6
444	M02	32486.3	19404.6	0.00	6
445	M02	33807.3	19404.6	0.00	6
446	M02	35128.3	19404.6	0.00	6
447	M02	36449.3	19404.6	0.00	6
448	M03	37770.3	19404.6	0.00	6
449	M01	20597.3	21788.6	0.00	6
450	M02	21918.3	21788.6	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 451–468

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
451	M02	23239.3	21788.6	0.00	6
452	M02	24560.3	21788.6	0.00	6
453	M02	25881.3	21788.6	0.00	6
454	M02	27202.3	21788.6	0.00	6
455	M02	28523.3	21788.6	0.00	6
456	M02	29844.3	21788.6	0.00	6
457	M02	31165.3	21788.6	0.00	6
458	M02	32486.3	21788.6	0.00	6
459	M02	33807.3	21788.6	0.00	6
460	M02	35128.3	21788.6	0.00	6
461	M02	36449.3	21788.6	0.00	6
462	M03	37770.3	21788.6	0.00	6
463	M01	41091.6	19472.6	0.00	6
464	M02	42412.6	19472.6	0.00	6
465	M02	43733.6	19472.6	0.00	6
466	M02	45054.6	19472.6	0.00	6
467	M02	46375.6	19472.6	0.00	6
468	M02	47696.6	19472.6	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 469–486

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
469	M02	49017.6	19472.6	0.00	6
470	M02	50338.6	19472.6	0.00	6
471	M02	51659.6	19472.6	0.00	6
472	M02	52980.6	19472.6	0.00	6
473	M02	54301.6	19472.6	0.00	6
474	M02	55622.6	19472.6	0.00	6
475	M02	56943.6	19472.6	0.00	6
476	M03	58264.6	19472.6	0.00	6
477	M01	41091.6	21856.6	0.00	6
478	M02	42412.6	21856.6	0.00	6
479	M02	43733.6	21856.6	0.00	6
480	M02	45054.6	21856.6	0.00	6
481	M02	46375.6	21856.6	0.00	6
482	M02	47696.6	21856.6	0.00	6
483	M02	49017.6	21856.6	0.00	6
484	M02	50338.6	21856.6	0.00	6
485	M02	51659.6	21856.6	0.00	6
486	M02	52980.6	21856.6	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 487–504

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
487	M02	54301.6	21856.6	0.00	6
488	M02	55622.6	21856.6	0.00	6
489	M02	56943.6	21856.6	0.00	6
490	M03	58264.6	21856.6	0.00	6
491	M01	61585.8	19472.6	0.00	6
492	M02	62906.8	19472.6	0.00	6
493	M02	64227.8	19472.6	0.00	6
494	M02	65548.8	19472.6	0.00	6
495	M02	66869.8	19472.6	0.00	6
496	M02	68190.8	19472.6	0.00	6
497	M02	69511.8	19472.6	0.00	6
498	M02	70832.8	19472.6	0.00	6
499	M02	72153.8	19472.6	0.00	6
500	M02	73474.8	19472.6	0.00	6
501	M02	74795.8	19472.6	0.00	6
502	M02	76116.8	19472.6	0.00	6
503	M03	77437.8	19472.6	0.00	6
504	M01	61585.8	21856.6	0.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 505–522

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
505	M02	62906.8	21856.6	0.00	6
506	M02	64227.8	21856.6	0.00	6
507	M02	65548.8	21856.6	0.00	6
508	M02	66869.8	21856.6	0.00	6
509	M02	68190.8	21856.6	0.00	6
510	M02	69511.8	21856.6	0.00	6
511	M02	70832.8	21856.6	0.00	6
512	M02	72153.8	21856.6	0.00	6
513	M02	73474.8	21856.6	0.00	6
514	M02	74795.8	21856.6	0.00	6
515	M02	76116.8	21856.6	0.00	6
516	M03	77437.8	21856.6	0.00	6
517	M04	70222.5	26549.1	90.00	6
518	M04	67838.5	26549.1	90.00	6
519	M04	65454.5	26549.1	90.00	6
520	M04	63070.5	26549.1	90.00	6
521	M04	60686.5	26549.1	90.00	6
522	M04	58302.5	26549.1	90.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / placement 523–533

Module	Detail	Corner 1 X mm	Corner 1 Y mm	Angle °	Points
523	M04	55918.5	26549.1	90.00	6
524	M04	53534.5	26549.1	90.00	6
525	M04	51150.5	26549.1	90.00	6
526	M04	48766.5	26549.1	90.00	6
527	M04	46382.5	26549.1	90.00	6
528	M04	43998.5	26549.1	90.00	6
529	M04	41614.5	26549.1	90.00	6
530	M04	39230.5	26549.1	90.00	6
531	M04	36846.5	26549.1	90.00	6
532	M04	34462.5	26549.1	90.00	6
533	M04	32078.5	26549.1	90.00	6

Datum O: TEMPORARY. Calibrated drawing coordinates: X rightwards, Y downwards. Angle from corner 1 to corner 2 is clockwise positive. Use the general drawing datum dimensions for positioning on the roof.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
1	P2	18.0	P16	0.0
2	P3	18.0	P17	0.0
3	P4	18.0	P18	0.0
4	P5	18.0	P19	0.0
5	P6	18.0	P20	0.0
6	P7	18.0	P21	0.0
7	P8	18.0	P22	0.0
8	P9	18.0	P23	0.0
9	P10	18.0	P24	0.0
10	P11	18.0	P25	0.0
11	P12	18.0	P26	0.0
12	P13	18.0	P27	0.0
13	P14	18.0	P28	0.0
14	P15	18.0	P29	0.0
15	P106	2000.0	P30	0.0
16	P17	18.0	P31	0.0
17	P18	18.0	P32	0.0
18	P19	18.0	P33	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
19	P20	18.0	P34	0.0
20	P21	18.0	P35	0.0
21	P22	18.0	P36	0.0
22	P23	18.0	P37	0.0
23	P24	18.0	P38	0.0
24	P25	18.0	P39	0.0
25	P26	18.0	P40	0.0
26	P27	18.0	P41	0.0
27	P28	18.0	P42	0.0
28	P29	18.0	P43	0.0
29	P30	18.0	P44	0.0
30	P121	2000.0	P45	0.0
31	P32	18.0	P46	0.0
32	P33	18.0	P47	0.0
33	P34	18.0	P48	0.0
34	P35	18.0	P49	0.0
35	P36	18.0	P50	0.0
36	P37	18.0	P51	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
37	P38	18.0	P52	0.0
38	P39	18.0	P53	0.0
39	P40	18.0	P54	0.0
40	P41	18.0	P55	0.0
41	P42	18.0	P56	0.0
42	P43	18.0	P57	0.0
43	P44	18.0	P58	0.0
44	P45	18.0	P59	0.0
45	P136	2000.0	P60	0.0
46	P47	18.0	P61	0.0
47	P48	18.0	P62	0.0
48	P49	18.0	P63	0.0
49	P50	18.0	P64	0.0
50	P51	18.0	P65	0.0
51	P52	18.0	P66	0.0
52	P53	18.0	P67	0.0
53	P54	18.0	P68	0.0
54	P55	18.0	P69	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
55	P56	18.0	P70	0.0
56	P57	18.0	P71	0.0
57	P58	18.0	P72	0.0
58	P59	18.0	P73	0.0
59	P60	18.0	P74	0.0
60	P151	2000.0	P75	0.0
61	P62	18.0	P76	0.0
62	P63	18.0	P77	0.0
63	P64	18.0	P78	0.0
64	P65	18.0	P79	0.0
65	P66	18.0	P80	0.0
66	P67	18.0	P81	0.0
67	P68	18.0	P82	0.0
68	P69	18.0	P83	0.0
69	P70	18.0	P84	0.0
70	P71	18.0	P85	0.0
71	P72	18.0	P86	0.0
72	P73	18.0	P87	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
73	P74	18.0	P88	0.0
74	P75	18.0	P89	0.0
75	P166	2000.0	P90	0.0
76	P77	18.0	P91	0.0
77	P78	18.0	P92	0.0
78	P79	18.0	P93	0.0
79	P80	18.0	P94	0.0
80	P81	18.0	P95	0.0
81	P82	18.0	P96	0.0
82	P83	18.0	P97	0.0
83	P84	18.0	P98	0.0
84	P85	18.0	P99	0.0
85	P86	18.0	P100	0.0
86	P87	18.0	P101	0.0
87	P88	18.0	P102	0.0
88	P89	18.0	P103	0.0
89	P90	18.0	P104	0.0
90	P181	2000.0	P105	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
91	P92	18.0	P386	1531.6
92	P93	18.0	P387	1531.6
93	P94	18.0	P388	1531.6
94	P95	18.0	P389	1531.6
95	P96	18.0	P390	1531.6
96	P97	18.0	P391	1531.6
97	P98	18.0	P402	6299.6
98	P99	18.0	P403	6299.6
99	P100	18.0	P404	6299.6
100	P101	18.0	P397	3915.6
101	P102	18.0	P398	3915.6
102	P103	18.0	P399	3915.6
103	P104	18.0	P392	1531.6
104	P105	18.0	P393	1531.6
105	P196	2000.0	-	-
106	P107	18.0	P121	0.0
107	P108	18.0	P122	0.0
108	P109	18.0	P123	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
109	P110	18.0	P124	0.0
110	P111	18.0	P125	0.0
111	P112	18.0	P126	0.0
112	P113	18.0	P127	0.0
113	P114	18.0	P128	0.0
114	P115	18.0	P129	0.0
115	P116	18.0	P130	0.0
116	P117	18.0	P131	0.0
117	P118	18.0	P132	0.0
118	P119	18.0	P133	0.0
119	P120	18.0	P134	0.0
120	P211	2000.0	P135	0.0
121	P122	18.0	P136	0.0
122	P123	18.0	P137	0.0
123	P124	18.0	P138	0.0
124	P125	18.0	P139	0.0
125	P126	18.0	P140	0.0
126	P127	18.0	P141	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
127	P128	18.0	P142	0.0
128	P129	18.0	P143	0.0
129	P130	18.0	P144	0.0
130	P131	18.0	P145	0.0
131	P132	18.0	P146	0.0
132	P133	18.0	P147	0.0
133	P134	18.0	P148	0.0
134	P135	18.0	P149	0.0
135	P226	2000.0	P150	0.0
136	P137	18.0	P151	0.0
137	P138	18.0	P152	0.0
138	P139	18.0	P153	0.0
139	P140	18.0	P154	0.0
140	P141	18.0	P155	0.0
141	P142	18.0	P156	0.0
142	P143	18.0	P157	0.0
143	P144	18.0	P158	0.0
144	P145	18.0	P159	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
145	P146	18.0	P160	0.0
146	P147	18.0	P161	0.0
147	P148	18.0	P162	0.0
148	P149	18.0	P163	0.0
149	P150	18.0	P164	0.0
150	P241	2000.0	P165	0.0
151	P152	18.0	P166	0.0
152	P153	18.0	P167	0.0
153	P154	18.0	P168	0.0
154	P155	18.0	P169	0.0
155	P156	18.0	P170	0.0
156	P157	18.0	P171	0.0
157	P158	18.0	P172	0.0
158	P159	18.0	P173	0.0
159	P160	18.0	P174	0.0
160	P161	18.0	P175	0.0
161	P162	18.0	P176	0.0
162	P163	18.0	P177	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
163	P164	18.0	P178	0.0
164	P165	18.0	P179	0.0
165	P256	2000.0	P180	0.0
166	P167	18.0	P181	0.0
167	P168	18.0	P182	0.0
168	P169	18.0	P183	0.0
169	P170	18.0	P184	0.0
170	P171	18.0	P185	0.0
171	P172	18.0	P186	0.0
172	P173	18.0	P187	0.0
173	P174	18.0	P188	0.0
174	P175	18.0	P189	0.0
175	P176	18.0	P190	0.0
176	P177	18.0	P191	0.0
177	P178	18.0	P192	0.0
178	P179	18.0	P193	0.0
179	P180	18.0	P194	0.0
180	P271	2000.0	P195	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
181	P182	18.0	P196	0.0
182	P183	18.0	P197	0.0
183	P184	18.0	P198	0.0
184	P185	18.0	P199	0.0
185	P186	18.0	P200	0.0
186	P187	18.0	P201	0.0
187	P188	18.0	P202	0.0
188	P189	18.0	P203	0.0
189	P190	18.0	P204	0.0
190	P191	18.0	P205	0.0
191	P192	18.0	P206	0.0
192	P193	18.0	P207	0.0
193	P194	18.0	P208	0.0
194	P195	18.0	P209	0.0
195	P286	2000.0	P210	0.0
196	P197	18.0	P436	2716.6
197	P198	18.0	P436	2716.6
198	P199	18.0	P437	2716.6

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
199	P200	18.0	P438	2716.6
200	P201	18.0	P439	2716.6
201	P202	18.0	P440	2716.6
202	P203	18.0	P441	2716.6
203	P204	18.0	P442	2716.6
204	P205	18.0	P443	2716.6
205	P206	18.0	P444	2716.6
206	P207	18.0	P445	2716.6
207	P208	18.0	P446	2716.6
208	P209	18.0	P447	2716.6
209	P210	18.0	P448	2716.6
210	P301	2000.0	P463	2784.6
211	P212	18.0	P226	0.0
212	P213	18.0	P227	0.0
213	P214	18.0	P228	0.0
214	P215	18.0	P229	0.0
215	P216	18.0	P230	0.0
216	P217	18.0	P231	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
217	P218	18.0	P232	0.0
218	P219	18.0	P233	0.0
219	P220	18.0	P234	0.0
220	P221	18.0	P235	0.0
221	P222	18.0	P236	0.0
222	P223	18.0	P237	0.0
223	P224	18.0	P238	0.0
224	P225	18.0	P239	0.0
225	P316	2000.0	P240	0.0
226	P227	18.0	P241	0.0
227	P228	18.0	P242	0.0
228	P229	18.0	P243	0.0
229	P230	18.0	P244	0.0
230	P231	18.0	P245	0.0
231	P232	18.0	P246	0.0
232	P233	18.0	P247	0.0
233	P234	18.0	P248	0.0
234	P235	18.0	P249	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
235	P236	18.0	P250	0.0
236	P237	18.0	P251	0.0
237	P238	18.0	P252	0.0
238	P239	18.0	P253	0.0
239	P240	18.0	P254	0.0
240	P326	2000.0	P255	0.0
241	P242	18.0	P256	0.0
242	P243	18.0	P257	0.0
243	P244	18.0	P258	0.0
244	P245	18.0	P259	0.0
245	P246	18.0	P260	0.0
246	P247	18.0	P261	0.0
247	P248	18.0	P262	0.0
248	P249	18.0	P263	0.0
249	P250	18.0	P264	0.0
250	P251	18.0	P265	0.0
251	P252	18.0	P266	0.0
252	P253	18.0	P267	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
253	P254	18.0	P268	0.0
254	P255	18.0	P269	0.0
255	P336	2000.0	P270	0.0
256	P257	18.0	P271	0.0
257	P258	18.0	P272	0.0
258	P259	18.0	P273	0.0
259	P260	18.0	P274	0.0
260	P261	18.0	P275	0.0
261	P262	18.0	P276	0.0
262	P263	18.0	P277	0.0
263	P264	18.0	P278	0.0
264	P265	18.0	P279	0.0
265	P266	18.0	P280	0.0
266	P267	18.0	P281	0.0
267	P268	18.0	P282	0.0
268	P269	18.0	P283	0.0
269	P270	18.0	P284	0.0
270	P346	2000.0	P285	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
271	P272	18.0	P286	0.0
272	P273	18.0	P287	0.0
273	P274	18.0	P288	0.0
274	P275	18.0	P289	0.0
275	P276	18.0	P290	0.0
276	P277	18.0	P291	0.0
277	P278	18.0	P292	0.0
278	P279	18.0	P293	0.0
279	P280	18.0	P294	0.0
280	P281	18.0	P295	0.0
281	P282	18.0	P296	0.0
282	P283	18.0	P297	0.0
283	P284	18.0	P298	0.0
284	P285	18.0	P299	0.0
285	P356	2000.0	P300	0.0
286	P287	18.0	P301	0.0
287	P288	18.0	P302	0.0
288	P289	18.0	P303	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
289	P290	18.0	P304	0.0
290	P291	18.0	P305	0.0
291	P292	18.0	P306	0.0
292	P293	18.0	P307	0.0
293	P294	18.0	P308	0.0
294	P295	18.0	P309	0.0
295	P296	18.0	P310	0.0
296	P297	18.0	P311	0.0
297	P298	18.0	P312	0.0
298	P299	18.0	P313	0.0
299	P300	18.0	P314	0.0
300	P366	2000.0	P315	0.0
301	P302	18.0	P464	2784.6
302	P303	18.0	P465	2784.6
303	P304	18.0	P466	2784.6
304	P305	18.0	P467	2784.6
305	P306	18.0	P468	2784.6
306	P307	18.0	P469	2784.6

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
307	P308	18.0	P470	2784.6
308	P309	18.0	P471	2784.6
309	P310	18.0	P472	2784.6
310	P311	18.0	P473	2784.6
311	P312	18.0	P474	2784.6
312	P313	18.0	P475	2784.6
313	P314	18.0	P476	2784.6
314	P315	18.0	P491	2784.6
315	P376	2000.0	P491	2784.6
316	P317	18.0	P326	0.0
317	P318	18.0	P327	0.0
318	P319	18.0	P328	0.0
319	P320	18.0	P329	0.0
320	P321	18.0	P330	0.0
321	P322	18.0	P331	0.0
322	P323	18.0	P332	0.0
323	P324	18.0	P333	0.0
324	P325	18.0	P334	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
325	-	-	P335	0.0
326	P327	18.0	P336	0.0
327	P328	18.0	P337	0.0
328	P329	18.0	P338	0.0
329	P330	18.0	P339	0.0
330	P331	18.0	P340	0.0
331	P332	18.0	P341	0.0
332	P333	18.0	P342	0.0
333	P334	18.0	P343	0.0
334	P335	18.0	P344	0.0
335	-	-	P345	0.0
336	P337	18.0	P346	0.0
337	P338	18.0	P347	0.0
338	P339	18.0	P348	0.0
339	P340	18.0	P349	0.0
340	P341	18.0	P350	0.0
341	P342	18.0	P351	0.0
342	P343	18.0	P352	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
343	P344	18.0	P353	0.0
344	P345	18.0	P354	0.0
345	-	-	P355	0.0
346	P347	18.0	P356	0.0
347	P348	18.0	P357	0.0
348	P349	18.0	P358	0.0
349	P350	18.0	P359	0.0
350	P351	18.0	P360	0.0
351	P352	18.0	P361	0.0
352	P353	18.0	P362	0.0
353	P354	18.0	P363	0.0
354	P355	18.0	P364	0.0
355	-	-	P365	0.0
356	P357	18.0	P366	0.0
357	P358	18.0	P367	0.0
358	P359	18.0	P368	0.0
359	P360	18.0	P369	0.0
360	P361	18.0	P370	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
361	P362	18.0	P371	0.0
362	P363	18.0	P372	0.0
363	P364	18.0	P373	0.0
364	P365	18.0	P374	0.0
365	-	-	P375	0.0
366	P367	18.0	P376	0.0
367	P368	18.0	P377	0.0
368	P369	18.0	P378	0.0
369	P370	18.0	P379	0.0
370	P371	18.0	P380	0.0
371	P372	18.0	P381	0.0
372	P373	18.0	P382	0.0
373	P374	18.0	P383	0.0
374	P375	18.0	P384	0.0
375	-	-	P385	0.0
376	P377	18.0	P493	2784.6
377	P378	18.0	P494	2784.6
378	P379	18.0	P495	2784.6

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
379	P380	18.0	P496	2784.6
380	P381	18.0	P497	2784.6
381	P382	18.0	P498	2784.6
382	P383	18.0	P499	2784.6
383	P384	18.0	P500	2784.6
384	P385	18.0	P501	2784.6
385	-	-	P502	2784.6
386	P387	18.0	P394	0.0
387	P388	18.0	P395	0.0
388	P389	18.0	P396	0.0
389	P390	18.0	P411	4768.0
390	P391	18.0	P412	4768.0
391	P392	7944.0	P413	4768.0
392	P393	18.0	P400	0.0
393	P436	3452.4	P401	0.0
394	P395	18.0	P422	4768.0
395	P396	18.0	P423	4768.0
396	P397	7944.0	P410	2384.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
397	P398	18.0	P405	0.0
398	P399	18.0	P406	0.0
399	P400	18.0	P407	0.0
400	P401	18.0	P408	0.0
401	P449	2131.4	P409	0.0
402	P403	18.0	P414	0.0
403	P404	18.0	P415	0.0
404	P405	18.0	P416	0.0
405	P406	18.0	P417	0.0
406	P407	18.0	P418	0.0
407	P408	18.0	P419	0.0
408	P409	18.0	P420	0.0
409	P449	2131.4	P421	0.0
410	P411	18.0	P424	0.0
411	P412	18.0	P425	0.0
412	P413	18.0	P426	0.0
413	P414	18.0	P427	0.0
414	P415	18.0	P428	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
415	P416	18.0	P429	0.0
416	P417	18.0	P430	0.0
417	P418	18.0	P431	0.0
418	P419	18.0	P432	0.0
419	P420	18.0	P433	0.0
420	P421	18.0	P434	0.0
421	-	-	P435	0.0
422	P423	18.0	-	-
423	P424	18.0	-	-
424	P425	18.0	-	-
425	P426	18.0	-	-
426	P427	18.0	-	-
427	P428	18.0	-	-
428	P429	18.0	-	-
429	P430	18.0	-	-
430	P431	18.0	-	-
431	P432	18.0	-	-
432	P433	18.0	-	-

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
433	P434	18.0	-	-
434	P435	18.0	-	-
435	-	-	-	-
436	P437	18.0	P450	0.0
437	P438	18.0	P451	0.0
438	P439	18.0	P452	0.0
439	P440	18.0	P453	0.0
440	P441	18.0	P454	0.0
441	P442	18.0	P455	0.0
442	P443	18.0	P456	0.0
443	P444	18.0	P457	0.0
444	P445	18.0	P458	0.0
445	P446	18.0	P459	0.0
446	P447	18.0	P460	0.0
447	P448	18.0	P461	0.0
448	P463	2018.2	P462	0.0
449	P450	18.0	-	-
450	P451	18.0	-	-

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
451	P452	18.0	-	-
452	P453	18.0	-	-
453	P454	18.0	-	-
454	P455	18.0	-	-
455	P456	18.0	-	-
456	P457	18.0	-	-
457	P458	18.0	-	-
458	P459	18.0	-	-
459	P460	18.0	-	-
460	P461	18.0	-	-
461	P462	18.0	-	-
462	P463	2018.2	-	-
463	P464	18.0	P477	0.0
464	P465	18.0	P478	0.0
465	P466	18.0	P479	0.0
466	P467	18.0	P480	0.0
467	P468	18.0	P481	0.0
468	P469	18.0	P482	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
469	P470	18.0	P483	0.0
470	P471	18.0	P484	0.0
471	P472	18.0	P485	0.0
472	P473	18.0	P486	0.0
473	P474	18.0	P487	0.0
474	P475	18.0	P488	0.0
475	P476	18.0	P489	0.0
476	P491	2018.2	P490	0.0
477	P478	18.0	-	-
478	P479	18.0	-	-
479	P480	18.0	-	-
480	P481	18.0	-	-
481	P482	18.0	-	-
482	P483	18.0	-	-
483	P484	18.0	-	-
484	P485	18.0	-	-
485	P486	18.0	-	-
486	P487	18.0	-	-

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
487	P488	18.0	-	-
488	P489	18.0	-	-
489	P490	18.0	-	-
490	P504	2018.2	-	-
491	P492	18.0	P504	0.0
492	P493	18.0	P505	0.0
493	P494	18.0	P506	0.0
494	P495	18.0	P507	0.0
495	P496	18.0	P508	0.0
496	P497	18.0	P509	0.0
497	P498	18.0	P510	0.0
498	P499	18.0	P511	0.0
499	P500	18.0	P512	0.0
500	P501	18.0	P513	0.0
501	P502	18.0	P514	0.0
502	P503	18.0	P515	0.0
503	-	-	P516	0.0
504	P505	18.0	-	-

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
505	P506	18.0	-	-
506	P507	18.0	-	-
507	P508	18.0	-	-
508	P509	18.0	-	-
509	P510	18.0	-	-
510	P511	18.0	-	-
511	P512	18.0	-	-
512	P513	18.0	-	-
513	P514	18.0	-	-
514	P515	18.0	-	-
515	P516	18.0	-	-
516	-	-	-	-
517	-	-	P518	0.0
518	-	-	P519	0.0
519	-	-	P520	0.0
520	-	-	P521	0.0
521	-	-	P522	0.0
522	-	-	P523	0.0

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

Mounting / module gaps

Module	Neighbor +U	Gap U mm	Neighbor +V	Gap V mm
523	-	-	P524	0.0
524	-	-	P525	0.0
525	-	-	P526	0.0
526	-	-	P527	0.0
527	-	-	P528	0.0
528	-	-	P529	0.0
529	-	-	P530	0.0
530	-	-	P531	0.0
531	-	-	P532	0.0
532	-	-	P533	0.0
533	-	-	-	-

U: corner 1 to corner 2; V: corner 1 to corner 4. Clear gaps between aligned parallel edges, computed from Layout. -: no neighbor on that side.

String schedule | 1-16

Every string, assigned inverter and MPPT

String	Modules	DC kWp	Inverter	MPPT	Drawing colour
String 1	17	12.24	INV1	1	Cyan
String 2	17	12.24	INV1	2	Magenta
String 3	17	12.24	INV1	3	Cyan
String 4	17	12.24	INV1	4	Magenta
String 5	17	12.24	INV1	5	Cyan
String 6	17	12.24	INV1	6	Magenta
String 7	17	12.24	INV1	7	Cyan
String 8	17	12.24	INV1	8	Magenta
String 9	17	12.24	INV1	8	Cyan
String 10	17	12.24	INV1	9	Magenta
String 11	17	12.24	INV1	9	Cyan
String 12	16	11.52	INV1	10	Magenta
String 13	17	12.24	INV2	1	Cyan
String 14	17	12.24	INV2	2	Magenta
String 15	16	11.52	INV2	3	Cyan
String 16	17	12.24	INV2	4	Magenta

String schedule | 17-32

Every string, assigned inverter and MPPT

String	Modules	DC kWp	Inverter	MPPT	Drawing colour
String 17	16	11.52	INV2	5	Cyan
String 18	16	11.52	INV2	6	Magenta
String 19	16	11.52	INV2	7	Cyan
String 20	16	11.52	INV2	8	Magenta
String 21	17	12.24	INV3	1	Cyan
String 22	17	12.24	INV3	2	Magenta
String 23	17	12.24	INV3	3	Cyan
String 24	17	12.24	INV3	4	Magenta
String 25	17	12.24	INV3	5	Cyan
String 26	17	12.24	INV3	6	Magenta
String 27	16	11.52	INV3	7	Cyan
String 28	16	11.52	INV3	8	Magenta
String 29	16	11.52	INV3	9	Cyan
String 30	16	11.52	INV3	10	Magenta
String 31	16	11.52	INV3	10	Cyan
String 32	17	12.24	INV2	9	Magenta

Module connection order | 1-12

IDs follow the electrical connection order within each string

String	Module identifiers, in connection order
String 1	1, 16, 31, 46, 61, 76, 91, 92, 77, 62, 47, 32, 17, 2, 3, 18, 33
String 2	48, 63, 78, 93, 94, 79, 64, 49, 34, 19, 4, 5, 20, 35, 50, 65, 80
String 3	95, 96, 81, 66, 51, 36, 21, 6, 7, 22, 37, 52, 67, 82, 97, 98, 83
String 4	68, 53, 38, 23, 8, 9, 24, 39, 54, 69, 84, 99, 100, 85, 70, 55, 40
String 5	25, 10, 11, 26, 41, 56, 71, 86, 101, 102, 87, 72, 57, 42, 27, 12, 13
String 6	28, 43, 58, 73, 88, 103, 104, 89, 74, 59, 44, 29, 14, 15, 30, 45, 60
String 7	75, 90, 105, 196, 181, 166, 151, 136, 121, 106, 107, 122, 137, 152, 167, 182, 197
String 8	198, 183, 168, 153, 138, 123, 108, 109, 124, 139, 154, 169, 184, 199, 200, 185, 170
String 9	155, 140, 125, 110, 111, 126, 141, 156, 171, 186, 201, 202, 187, 172, 157, 142, 127
String 10	112, 113, 128, 143, 158, 173, 188, 203, 204, 189, 174, 159, 144, 129, 114, 115, 130
String 11	145, 160, 175, 190, 205, 206, 191, 176, 161, 146, 131, 116, 117, 132, 147, 162, 177
String 12	192, 207, 208, 193, 178, 163, 148, 133, 118, 119, 134, 149, 164, 179, 194, 209

Module connection order | 13-24

IDs follow the electrical connection order within each string

String	Module identifiers, in connection order
String 13	391, 390, 389, 388, 387, 386, 394, 395, 396, 410, 411, 412, 426, 425, 424, 423, 422
String 14	406, 398, 397, 405, 417, 431, 430, 416, 404, 403, 415, 429, 428, 414, 402, 413, 427
String 15	418, 432, 433, 419, 407, 399, 392, 400, 408, 420, 434, 435, 421, 409, 401, 393
String 16	449, 450, 436, 437, 451, 452, 438, 439, 453, 454, 440, 441, 455, 456, 442, 443, 457
String 17	458, 444, 445, 459, 460, 446, 447, 461, 462, 448, 463, 477, 478, 464, 465, 479
String 18	480, 466, 467, 481, 482, 468, 469, 483, 484, 470, 471, 485, 486, 472, 473, 487
String 19	488, 474, 475, 489, 490, 476, 491, 504, 505, 492, 493, 506, 507, 494, 495, 508
String 20	509, 496, 497, 510, 511, 498, 499, 512, 513, 500, 501, 514, 515, 502, 503, 516
String 21	120, 135, 150, 165, 180, 195, 210, 301, 286, 271, 256, 241, 226, 211, 212, 227, 242
String 22	257, 272, 287, 302, 303, 288, 273, 258, 243, 228, 213, 214, 229, 244, 259, 274, 289
String 23	304, 305, 290, 275, 260, 245, 230, 215, 216, 231, 246, 261, 276, 291, 306, 307, 292
String 24	277, 262, 247, 232, 217, 218, 233, 248, 263, 278, 293, 308, 309, 294, 279, 264, 249

Module connection order | 25-32

IDs follow the electrical connection order within each string

String	Module identifiers, in connection order
String 25	234, 219, 220, 235, 250, 265, 280, 295, 310, 311, 296, 281, 266, 251, 236, 221, 222
String 26	237, 252, 267, 282, 297, 312, 313, 298, 283, 268, 253, 238, 223, 224, 239, 254, 269
String 27	284, 299, 314, 315, 300, 285, 270, 255, 240, 225, 316, 326, 336, 346, 356, 366
String 28	376, 377, 367, 357, 347, 337, 327, 317, 318, 328, 338, 348, 358, 368, 378, 379
String 29	369, 359, 349, 339, 329, 319, 320, 330, 340, 350, 360, 370, 380, 381, 371, 361
String 30	351, 341, 331, 321, 322, 332, 342, 352, 362, 372, 382, 383, 373, 363, 353, 343
String 31	333, 323, 324, 334, 344, 354, 364, 374, 384, 385, 375, 365, 355, 345, 335, 325
String 32	517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533

Installation-zone dimensions

Scaled plan dimensions; mounting exclusions remain project inputs

Zone	Width m	Depth m	Area m2	Modules	Layout rotation deg	Cable level m
1	19.80	16.69	330.4	105	0.0	11.0
2	19.80	16.69	330.4	105	0.0	11.0
3	19.80	16.69	330.4	105	0.0	11.0
4	13.19	16.69	220.2	70	0.0	11.0
5	18.48	11.92	220.2	50	0.0	11.0
6	18.48	4.77	88.1	27	0.0	8.0
7	18.48	4.77	88.1	28	0.0	8.0
8	17.16	4.77	81.8	26	0.0	8.0
9	40.85	1.56	63.8	17	90.0	8.0

Layout rotation is the in-plane angular rotation of the module placement grid within the installation zone. It is not the module tilt relative to horizontal. The entered module tilt is 8 degrees and is recorded separately in the technical-input schedule. Roof elevation for shading and cable installation elevation are also separate inputs.

Site, solar and calculation assumptions

Entered values, displayed with human-readable units

Technical input	Value	Unit
Latitude	43.550	deg N
Longitude	10.310	deg E
Image scale	0.027	px/mm
Module tilt	8.000	deg
Module azimuth	180.000	deg
North correction	10.000	deg
NOCT	43.000	deg C
Temperature coefficient	-0.290	%/deg C
Obstacle height	18,000.000	mm
Obstacle width	3,000.000	mm
Obstacle opacity	0.450	fraction

Modules equipment schedule

Manufacturer and electrical records; evaluated results replace formulas | rows 1-17

Equipment row	Parameter	Value
1	Brand	Trina Solar
1	Model	TSM-NEG21C.20 (720 W)
1	Pmax (W)	720
1	Voc (V)	49.4
1	Isc (A)	18.49
1	Vmp (V)	41.3
1	Imp (A)	17.44
1	Rendement module (%)	23.2
1	Dimensions (mm)	2384 x 1303 x 33
1	Poids (kg)	38.3
1	NOCT (°C)	43
1	Coeff. Pmax (%/°C)	-0.29
1	Coeff. Voc (%/°C)	-0.24
1	Coeff. Isc (%/°C)	0.04
1	Tension système max (VDC)	1500
1	Fusible série max (A)	35
1	Bifacialité puissance (%)	80 ± 5

Inverters equipment schedule

Manufacturer and electrical records; evaluated results replace formulas | rows 1-24

Equipment row	Parameter	Value
1	Brand	Deye
1	Model	SUN-125K-SG02HP3-EU-GM10
1	Power (kVA)	125
1	Active power (kW)	125
1	MPPT inputs	10
1	Max voltage (V)	1000
1	Puissance apparente max (kVA)	135
1	Puissance PV accès max (kW)	250
1	Puissance PV entrée max (kW)	200
1	Tension démarrage PV (V)	180
1	Plage MPPT (V)	150-850
1	Tension PV nominale (V)	650
1	Courant MPPT max par tracker (A)	42
1	Courant Isc max par tracker (A)	63
1	Strings par MPPT	2
1	Plage batterie (V)	160-1000
1	Courant charge/décharge batterie max (A)	200
1	Entrées batterie	2
1	Rendement max (%)	98.7
1	Rendement euro (%)	98.1
1	Rendement MPPT (%)	>99
1	Tension AC nominale	220/380 V, 230/400 V
1	IP	IP65
1	SPD	Type II DC / Type II AC

Custom equipment schedule

Manufacturer and electrical records; evaluated results replace formulas | rows 1-40

Equipment row	Parameter	Value
1	Field	PV module datasheet
1	Value	Trina Solar TSM-NEG21C.20, 720 W bin
2	Field	PV module dimensions (mm)
2	Value	2384 x 1303 x 33
3	Field	PV module weight (kg)
3	Value	38.3
4	Field	PV module max system voltage (VDC)
4	Value	1500
5	Field	PV module max series fuse (A)
5	Value	35
6	Field	PV module power bifaciality
6	Value	80 ± 5 %
7	Field	Inverter datasheet
7	Value	Deye SUN-125K-SG02HP3-EU-GM10
8	Field	Inverter max PV access power (kW)
8	Value	250
9	Field	Inverter max PV input power (kW)
9	Value	200
10	Field	Inverter MPPT range (V)
10	Value	150-850
11	Field	Inverter max operating current / MPPT (A)
11	Value	42
12	Field	Inverter max short-circuit current / MPPT (A)
12	Value	63

Equipment row	Parameter	Value
13	Field	Inverter max efficiency (%)
13	Value	98.7
14	Field	BESS model
14	Value	Deye MC-L522-BC-2/3 / MC-L522-2H2/3 family
15	Field	BESS nominal energy (kWh)
15	Value	522.496
16	Field	BESS nominal DC voltage (V)
16	Value	832
17	Field	BESS DC voltage range (V)
17	Value	715-936
18	Field	BESS nominal capacity (Ah)
18	Value	314
19	Field	BESS charge/discharge rate
19	Value	0.5P
20	Field	BESS system topology
20	Value	2 battery clusters; 2 x SUN-125K hybrid inverters; PCS-BMS CAN / PCS parallel CAN

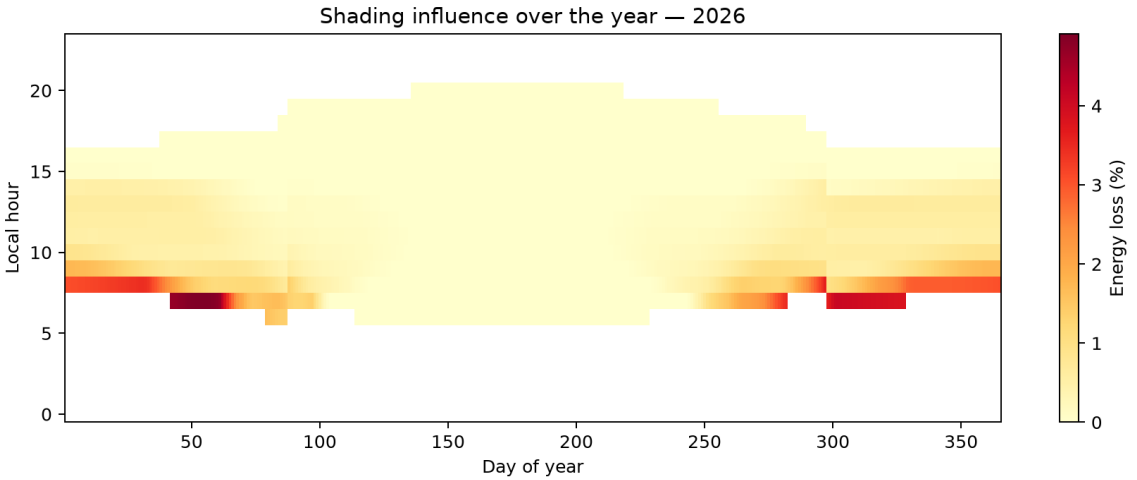
Annual shading influence

2026 | 8,760 elapsed hours | clear-sky geometric model

725,197.9 kWh	**723,834.0 kWh**	**1,363.9 kWh**	**0.188 %**
Unshaded potential	With shading	Shading loss	Annual loss

Method

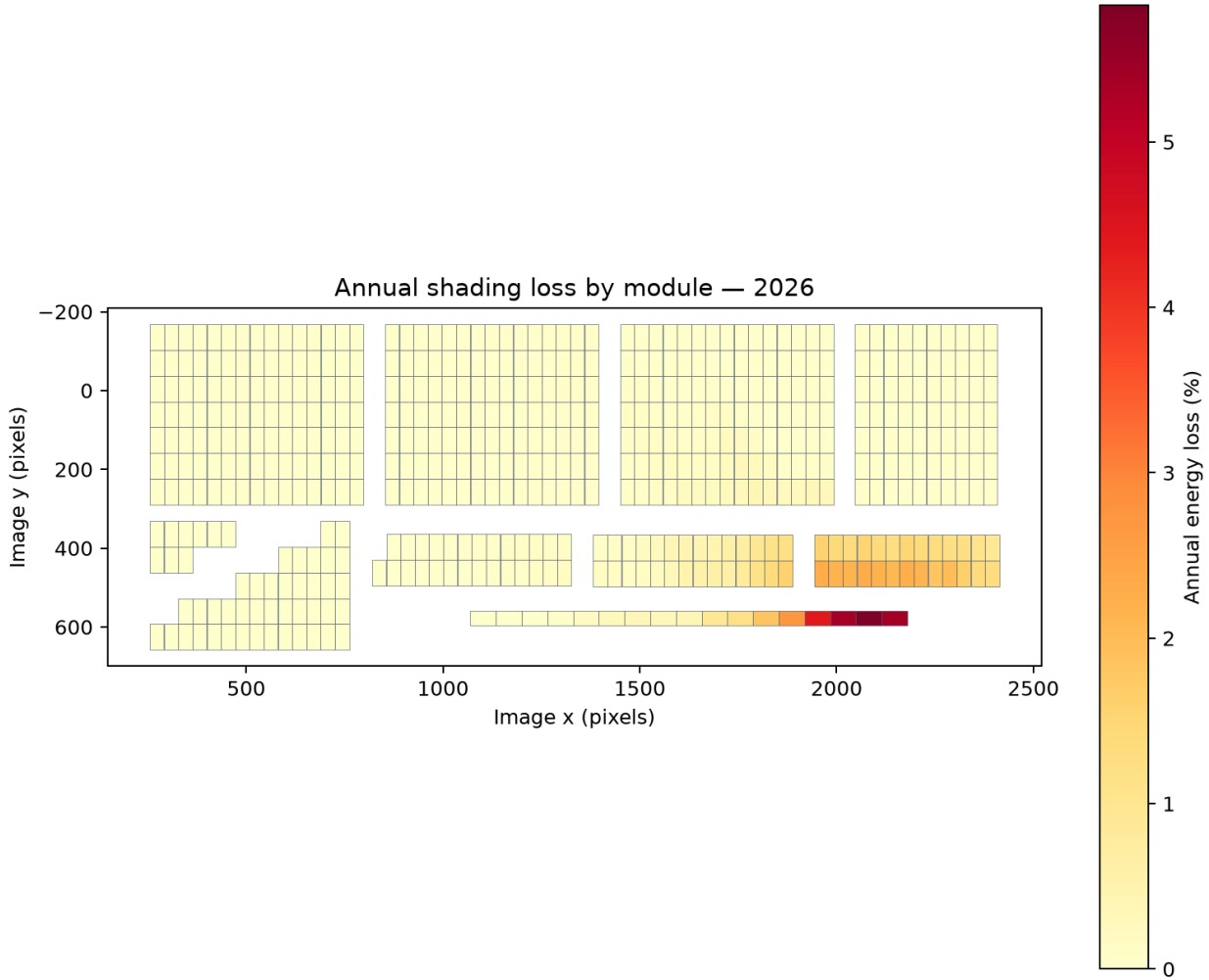
Integrate the current geometry, site orientation, thermal settings and obstacle model over the full calendar year. Shading loss is weighted by unshaded energy. This clear-sky study is distinct from the weather-driven energy/BESS calculation.



Daily/hourly shading-loss heatmap. Blank cells have no unshaded generation.

Spatial shading heatmap

Energy-weighted loss for each module



Warm colours indicate higher annual loss. See the colour bar for the numerical scale.

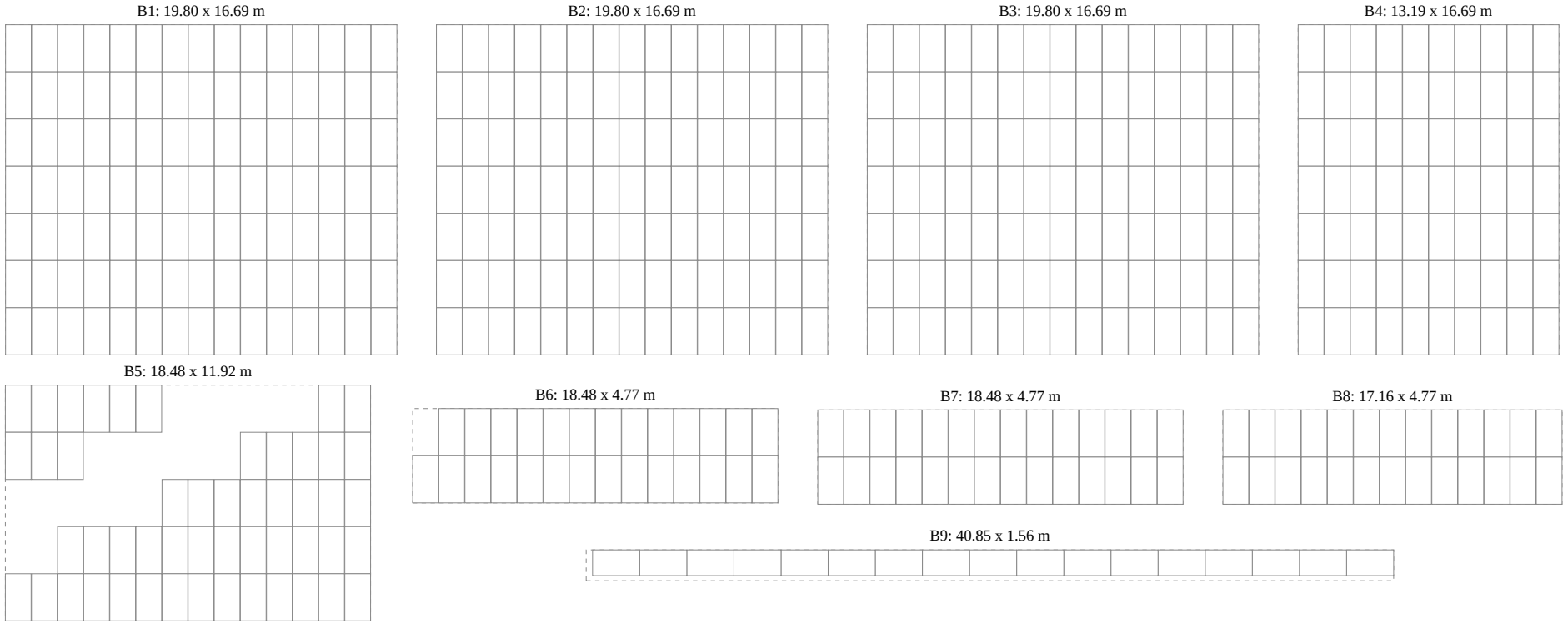
Monthly shading balance

A complete year, consistent with the heatmaps

Month	Unshaded kWh	Produced kWh	Lost kWh	Loss %
2026-01	28,013.2	27,827.5	185.7	0.663
2026-02	36,826.8	36,634.6	192.1	0.522
2026-03	58,914.4	58,742.9	171.5	0.291
2026-04	74,454.0	74,383.6	70.4	0.095
2026-05	89,782.2	89,774.7	7.5	0.008
2026-06	92,285.7	92,285.7	0.0	0.000
2026-07	93,059.4	93,059.1	0.2	0.000
2026-08	83,103.8	83,062.4	41.5	0.050
2026-09	64,881.2	64,765.2	116.0	0.179
2026-10	48,241.7	48,027.9	213.8	0.443
2026-11	31,101.5	30,913.0	188.5	0.606
2026-12	24,534.0	24,357.4	176.6	0.720

DC cable routing plan

Conductor routes and gathering locations



Routes are shown only where the required elevation inputs are available.

Per-string cable sizing | 1-14

One row per string, both conductor lengths and selected working section

String	A m	B m	Loop m	mm2	Drop %	Mass kg	Check status
String 1	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 2	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 3	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 4	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 5	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 6	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 7	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 8	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 9	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 10	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 11	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 12	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 13	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 14	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete

A and B are the two string-to-inverter conductors. Cross-section proposals must be checked against admissible current, installation conditions and protection coordination. Missing mass data remains explicit.

Per-string cable sizing | 15-28

One row per string, both conductor lengths and selected working section

String	A m	B m	Loop m	mm2	Drop %	Mass kg	Check status
String 15	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 16	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 17	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 18	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 19	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 20	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 21	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 22	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 23	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 24	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 25	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 26	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 27	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 28	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete

A and B are the two string-to-inverter conductors. Cross-section proposals must be checked against admissible current, installation conditions and protection coordination. Missing mass data remains explicit.

Per-string cable sizing | 29-32

One row per string, both conductor lengths and selected working section

String	A m	B m	Loop m	mm2	Drop %	Mass kg	Check status
String 29	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 30	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 31	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete
String 32	-	-	-	-	-	-	Unavailable: Route/elevation inputs incomplete

A and B are the two string-to-inverter conductors. Cross-section proposals must be checked against admissible current, installation conditions and protection coordination. Missing mass data remains explicit.

Energy and BESS comparison

Energy results current | clear-sky estimate + shading

Scenario	PV MWh	Self-use MWh	Grid MWh	To battery MWh	BESS loss MWh	Export MWh	Curtailment MWh
PV only	651.440	167.781	317.917	0.000	0.000	483.277	0.381
One BESS	651.440	310.299	175.400	158.468	15.841	324.810	0.381
Two BESS	651.440	423.159	62.539	284.309	28.404	199.101	0.249

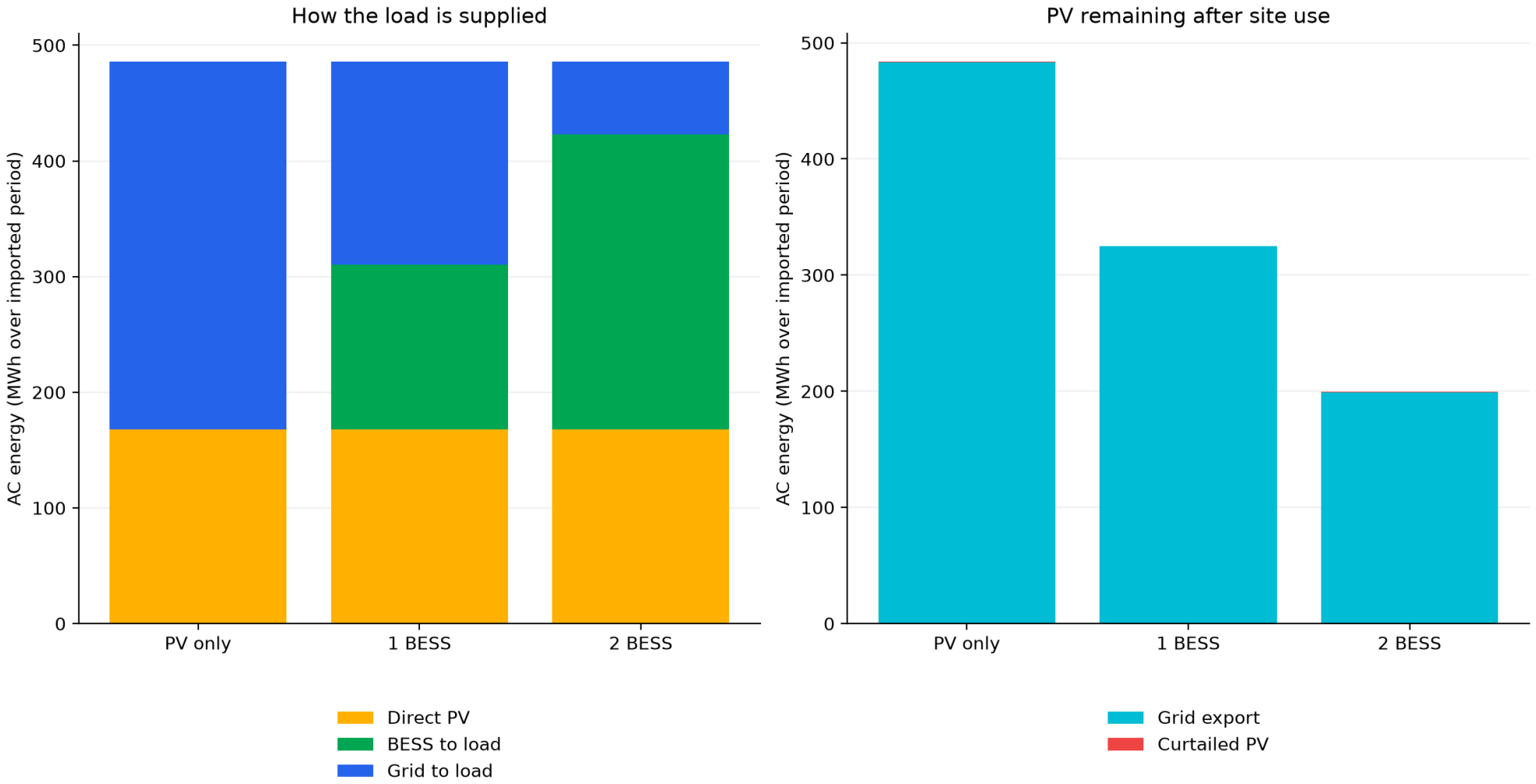
Monthly energy table

Current one-BESS result

Month	PV MWh	Load MWh	Export MWh
2025-04	66.9	41.7	38.3
2025-05	80.8	44.2	48.3
2025-06	83.0	48.6	47.9
2025-07	83.8	51.5	48.0
2025-08	74.8	48.2	41.6
2025-09	58.3	41.8	30.4
2025-10	43.2	39.9	16.9
2025-11	27.8	36.5	6.6
2025-12	21.9	34.2	2.6
2026-01	25.0	34.8	3.7
2026-02	33.0	30.4	12.3
2026-03	52.9	33.9	28.2

Energy supply and PV allocation

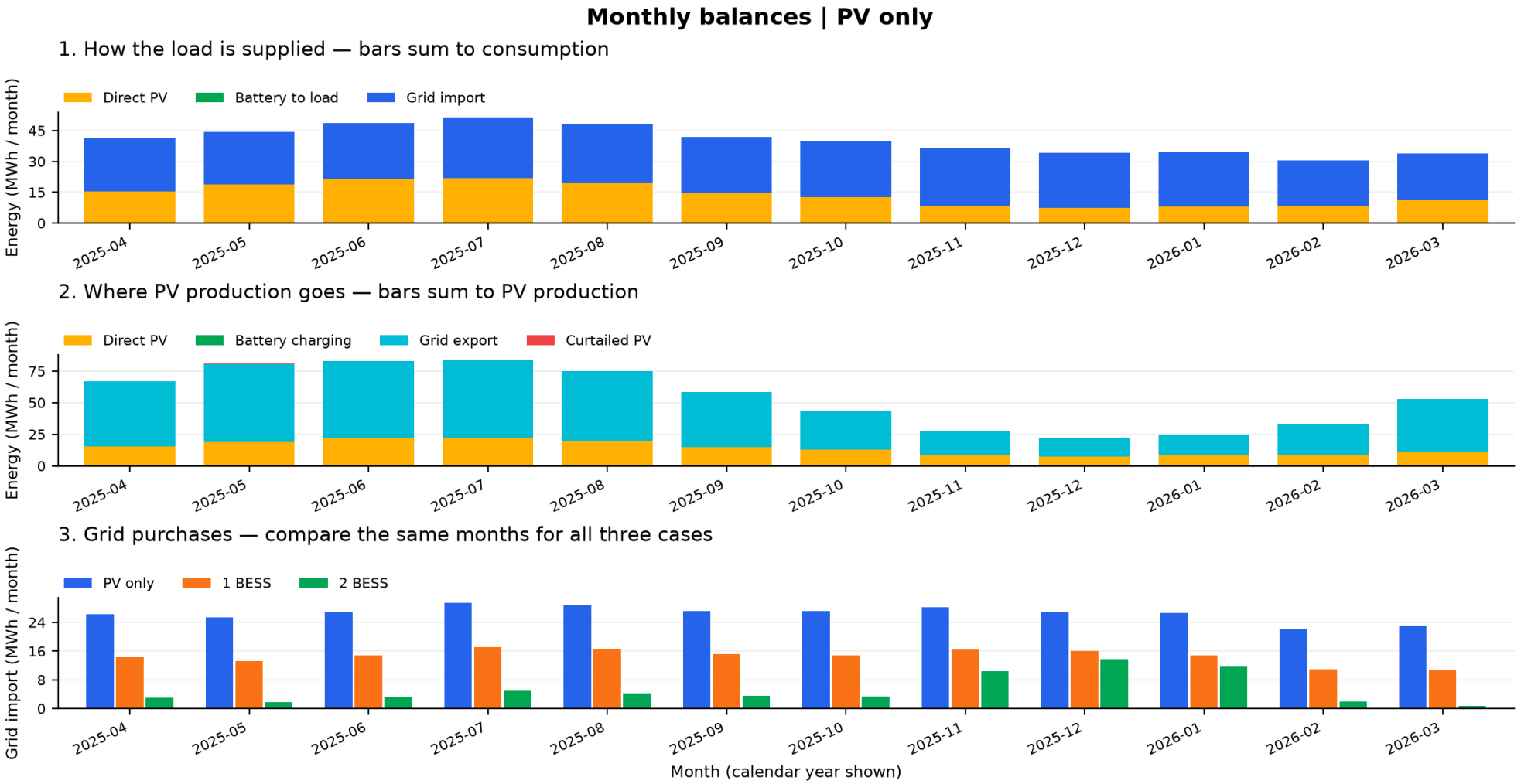
PV only / one BESS / two BESS | imported period: 2025-04-01 to 2026-03-31



Same comparison as the Energy tab. Totals cover the imported period, which may be shorter than a year.

Monthly energy balance | PV only

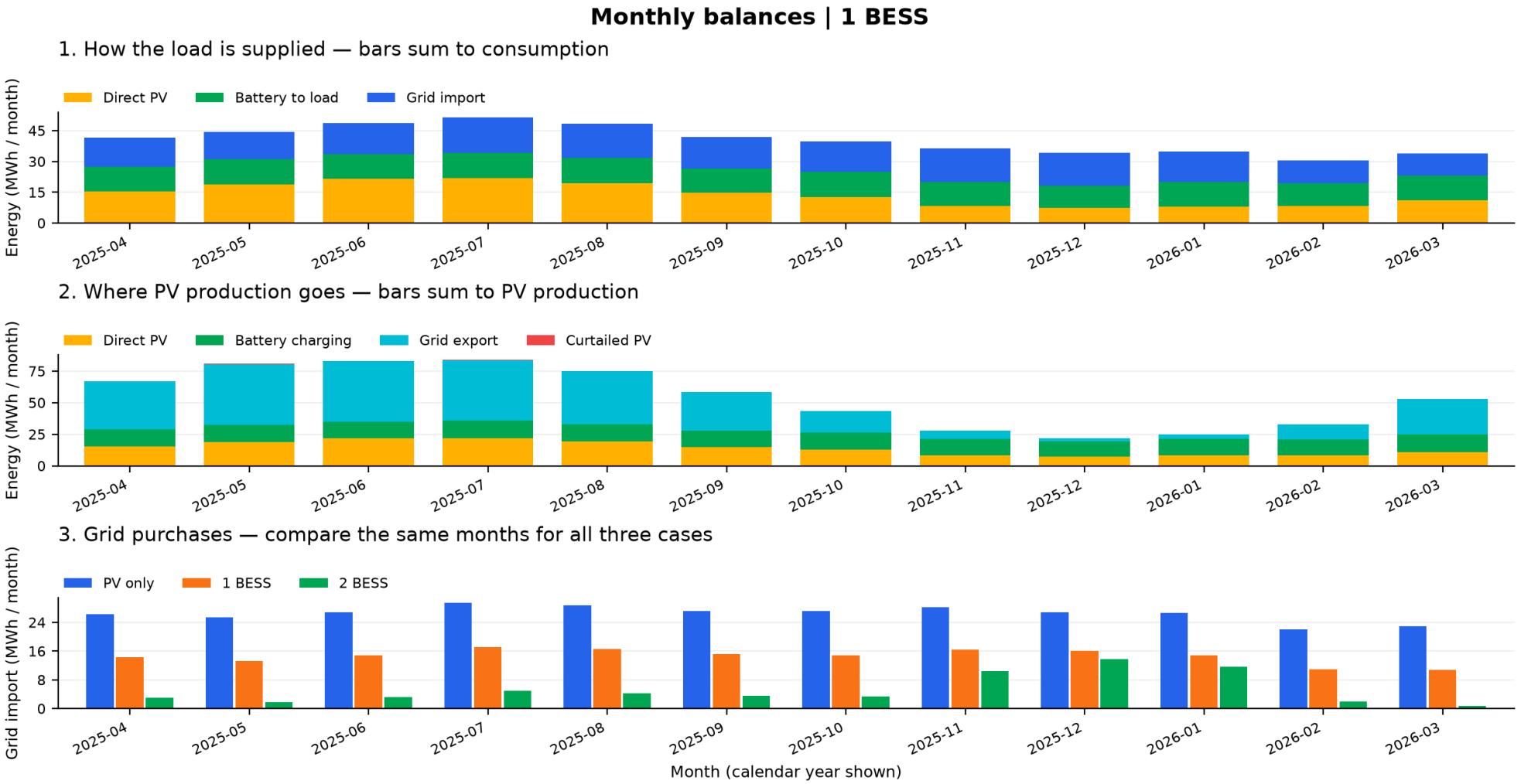
Imported period: 2025-04-01 to 2026-03-31



Load supply, PV allocation and grid purchases use the same axes, units and colours as the BESS monthly view.

Monthly energy balance | 1 BESS

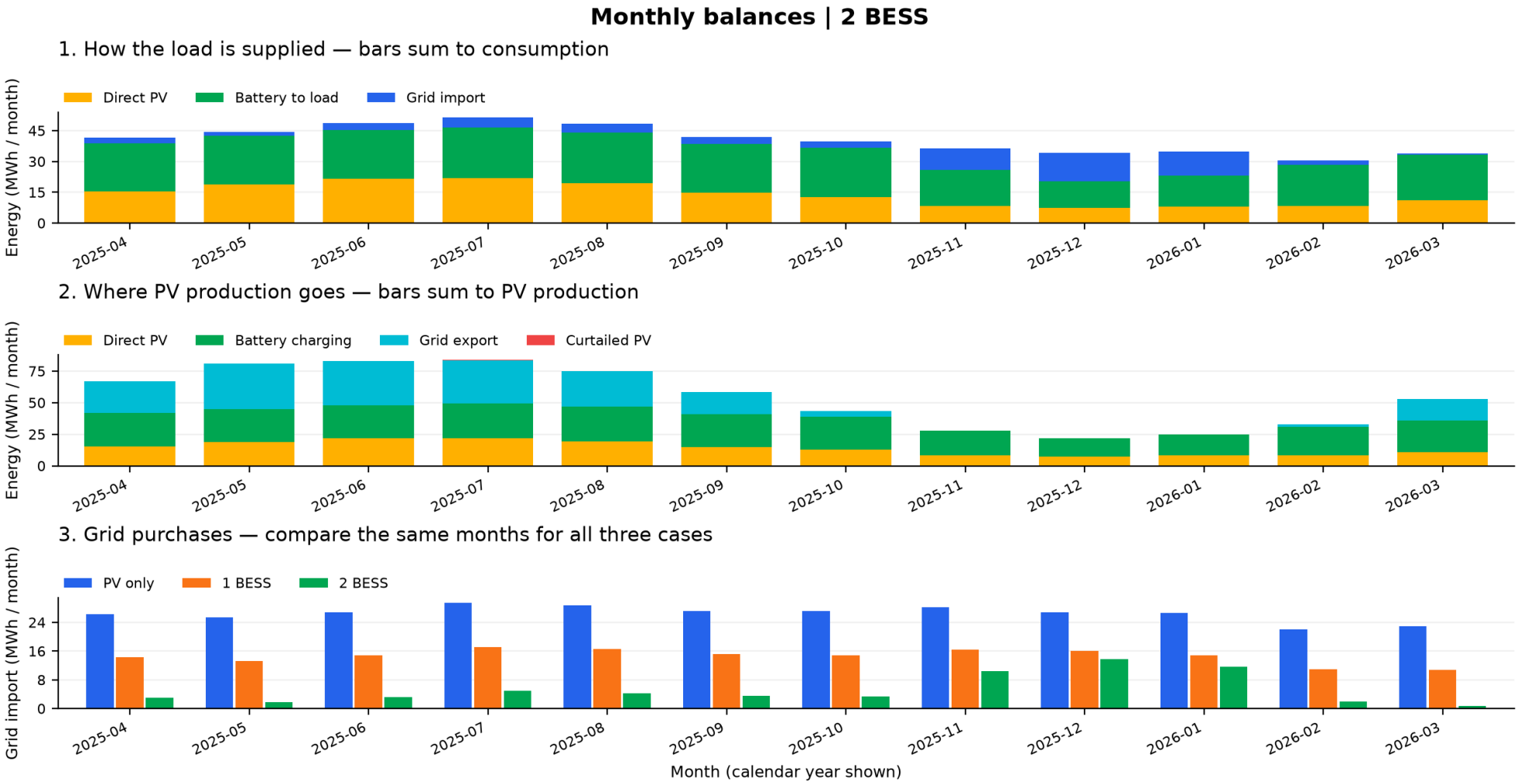
Imported period: 2025-04-01 to 2026-03-31



Load supply, PV allocation and grid purchases use the same axes, units and colours as the BESS monthly view.

Monthly energy balance | 2 BESS

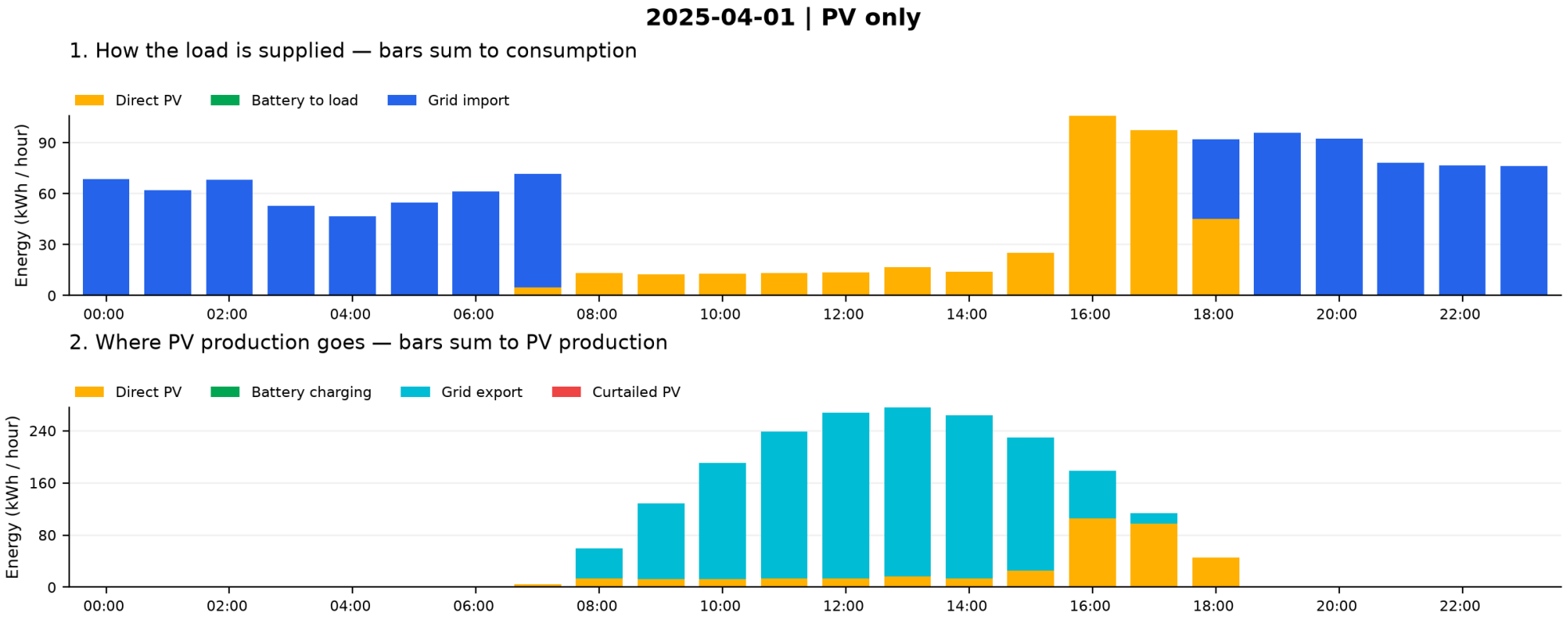
Imported period: 2025-04-01 to 2026-03-31



Load supply, PV allocation and grid purchases use the same axes, units and colours as the BESS monthly view.

Hourly energy balance | PV only

First imported local day: 2025-04-01 | 24 hourly intervals

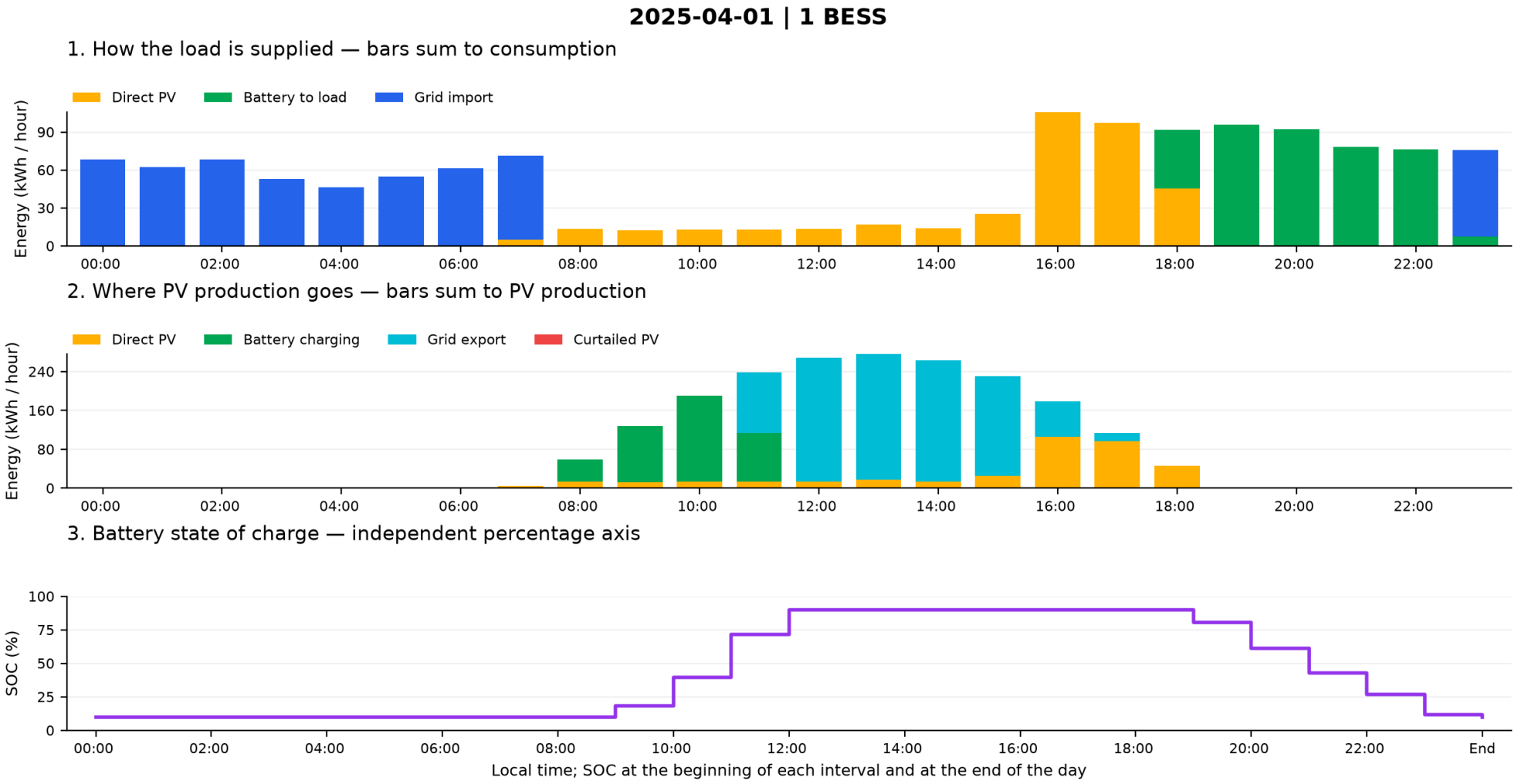


PV only: no battery charging, discharge or SOC.

Same daily view as the BESS tab. Battery charge/discharge are separate; SOC has its own percentage axis. This day does not represent the full imported period.

Hourly energy balance | 1 BESS

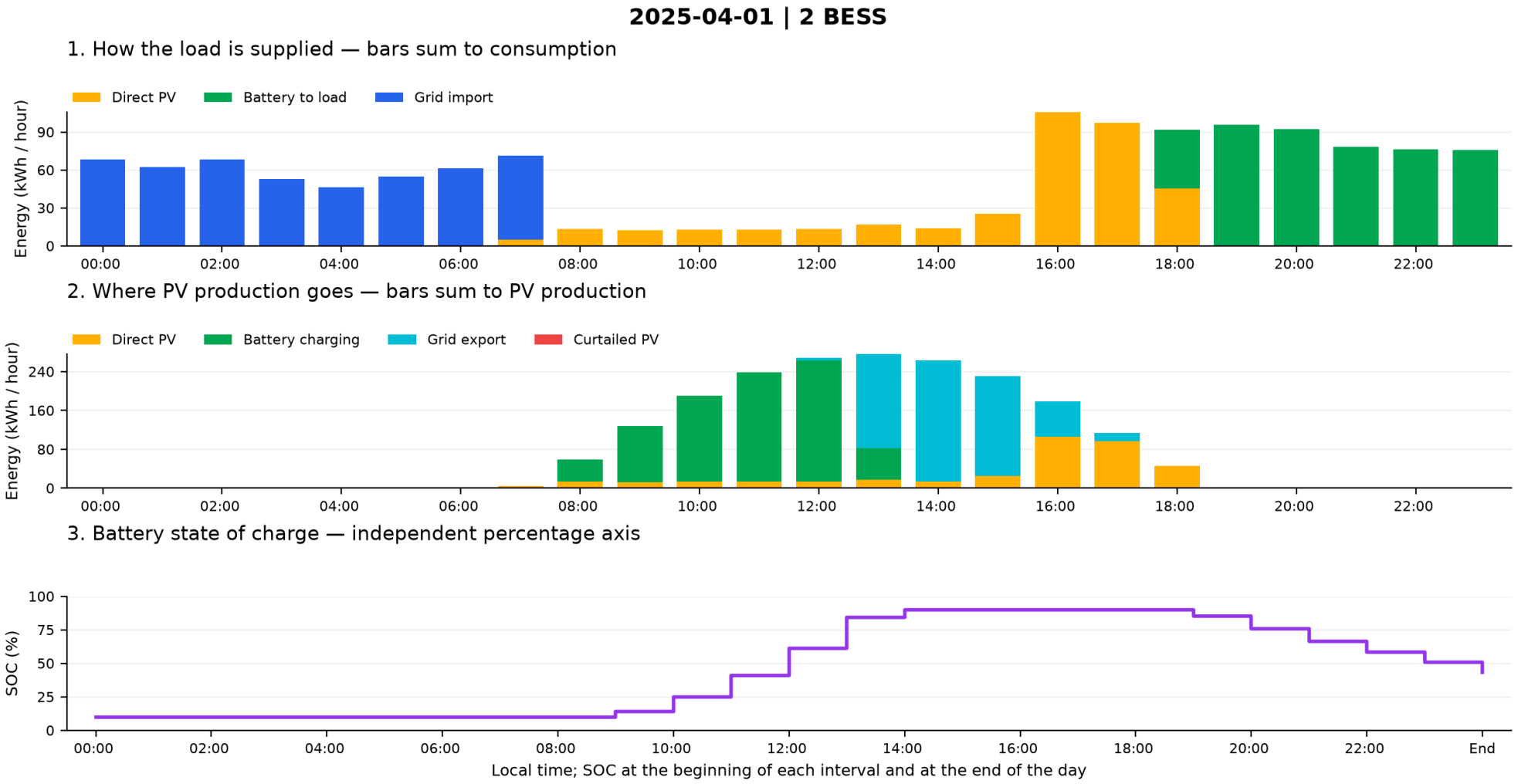
First imported local day: 2025-04-01 | 24 hourly intervals



Same daily view as the BESS tab. Battery charge/discharge are separate; SOC has its own percentage axis. This day does not represent the full imported period.

Hourly energy balance | 2 BESS

First imported local day: 2025-04-01 | 24 hourly intervals



Same daily view as the BESS tab. Battery charge/discharge are separate; SOC has its own percentage axis. This day does not represent the full imported period.

Economic comparison

Benefits cover the imported period; payback requires a complete year

BESS	Export EUR/kWh	Benefit EUR	Investment EUR	Payback years	NPV EUR
0	0.07	75,774.7805	250,000	3.2993	886,621.7069
1	0.07	100,298.9188	300,000	2.9911	1,204,483.782
2	0.07	119,702.0177	350,000	2.9239	1,445,530.2661
0	0.08	80,607.5551	250,000	3.1014	959,113.3267
1	0.08	103,547.0143	300,000	2.8972	1,253,205.2149
2	0.08	121,693.0258	350,000	2.8761	1,475,395.3875

Initial storage energy is excluded from the PV credit. Battery charging is not added to useful self-consumption.

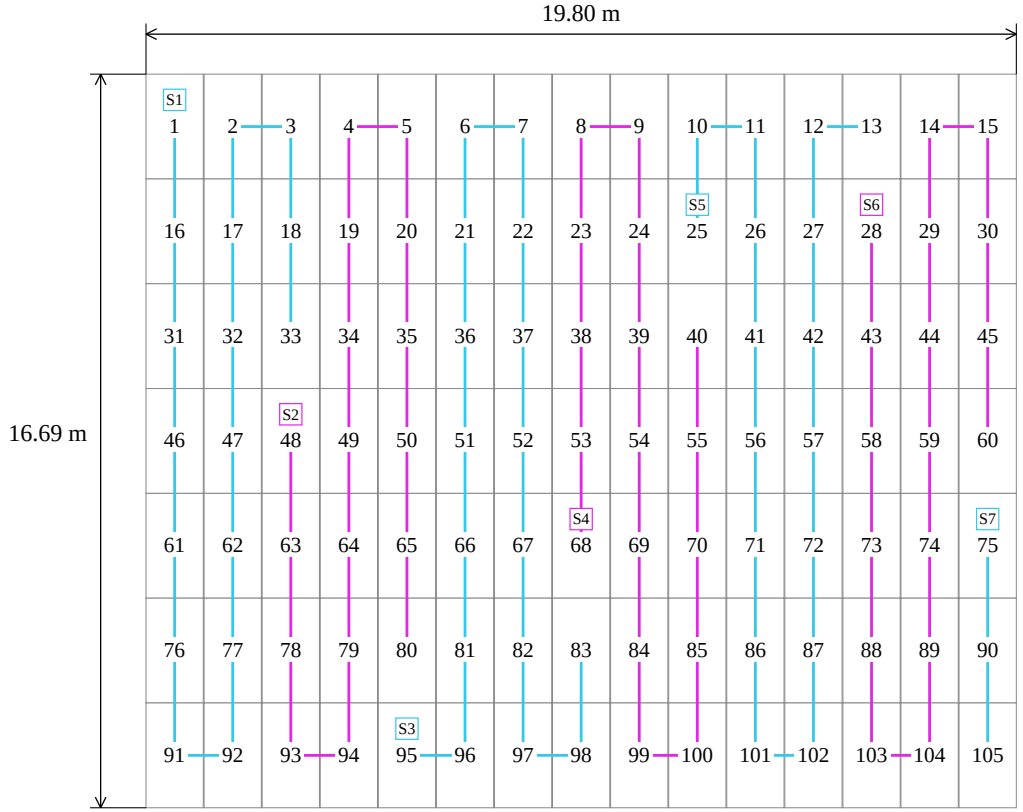
Complete spreadsheet | A-F | rows 1-15

Entire sheet on one page; formulas and evaluated values remain visible

Row	A	B	C	D	E	F
1	-	-	-	-	-	-
2	-	-	-	-	-	-
3	-	-	-	-	-	-
4	Weight B1	=B15*ZONE1_PANEL_COUNT Value: 4,021.5	kg	Power	=PV_CAPACITY_WP Value: 383,760	Wc
5	Weight B2	=B15*ZONE2_PANEL_COUNT Value: 4,021.5	kg	Project DC power	=PV_CAPACITY_WP Value: 383,760	Wc
6	Weight B3	=B15*ZONE3_PANEL_COUNT Value: 4,021.5	kg	-	-	-
7	Weight B4	=B15*ZONE4_PANEL_COUNT Value: 2,681	kg	Difference	=E4-E5 Value: 0	Wc
8	Weight B5	=B15*ZONE5_PANEL_COUNT Value: 1,915	kg	-	-	-
9	Weight B6	=B15*ZONE6_PANEL_COUNT Value: 1,034.1	kg	-	-	-
10	Weight B7	=B15*ZONE7_PANEL_COUNT Value: 1,072.4	kg	-	-	-
11	Weight B8	=B15*ZONE8_PANEL_COUNT Value: 995.8	kg	-	-	-
12	Weight B9	=ZONE9_PANEL_COUNT*B15 Value: 651.1	kg	-	-	-
13	Tot	=SUM(B4:B12) Value: 20,413.9	kg	-	-	-
14	-	-	-	-	-	-
15	1 pannel weight	38.3	kg	-	-	-

Physical block B1

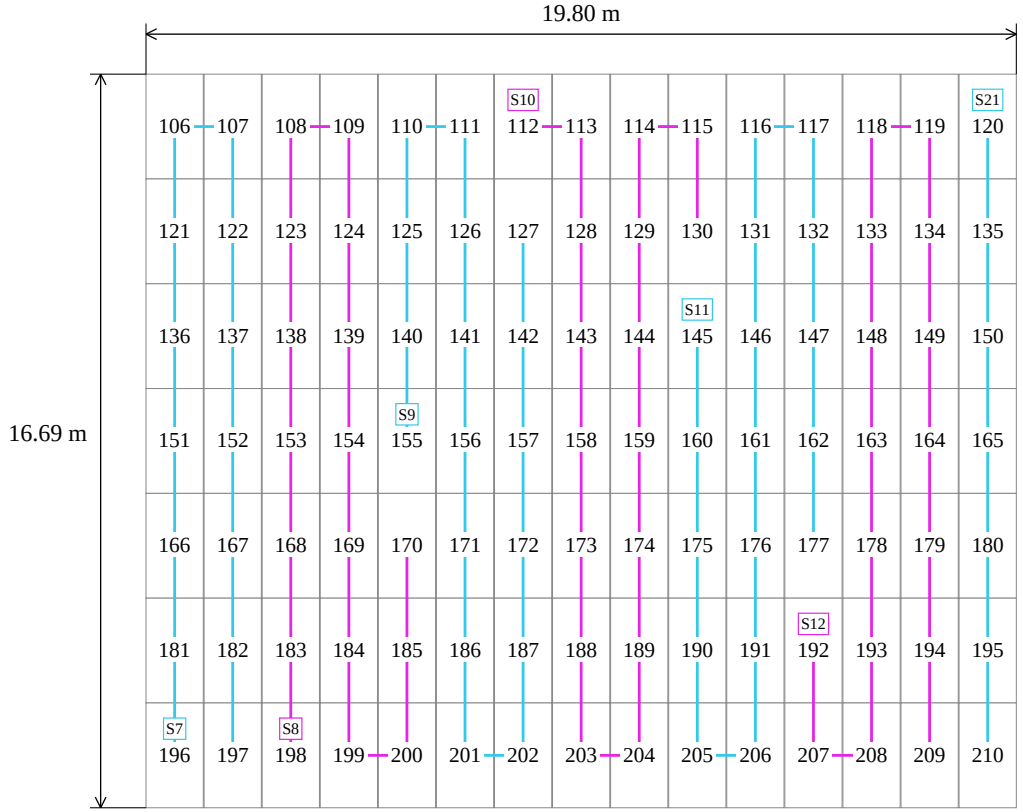
Zone 1 | 105 modules



Zone dimensions in metres; module IDs and string connections shown.

Physical block B2

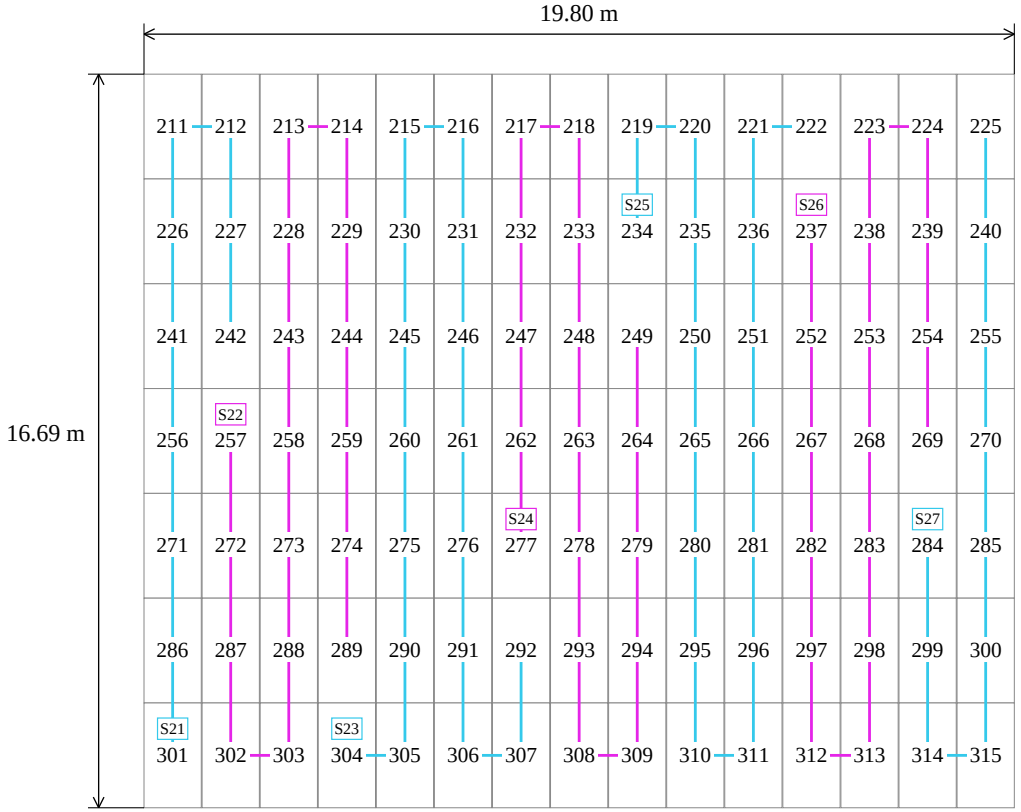
Zone 2 | 105 modules



Zone dimensions in metres; module IDs and string connections shown.

Physical block B3

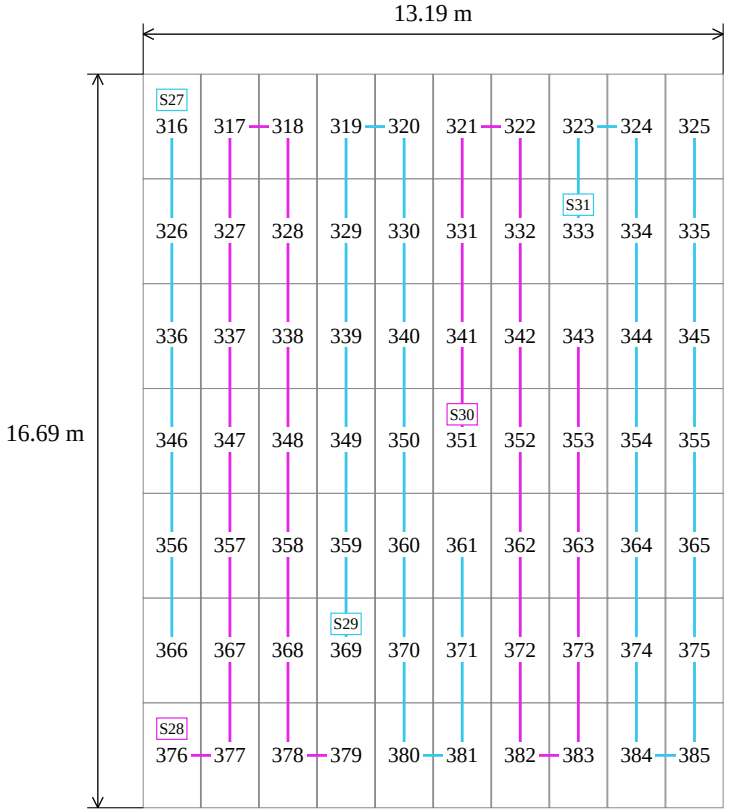
Zone 3 | 105 modules



Zone dimensions in metres; module IDs and string connections shown.

Physical block B4

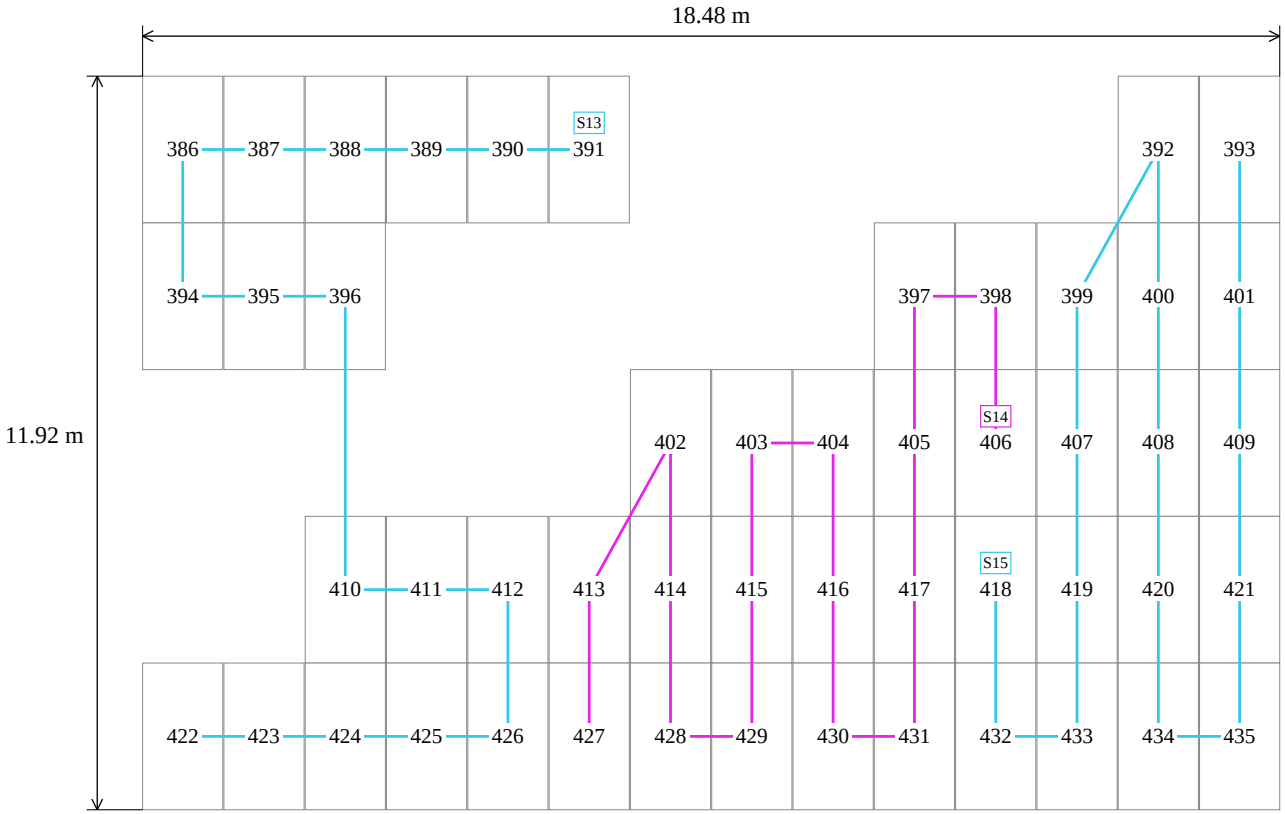
Zone 4 | 70 modules



Zone dimensions in metres; module IDs and string connections shown.

Physical block B5

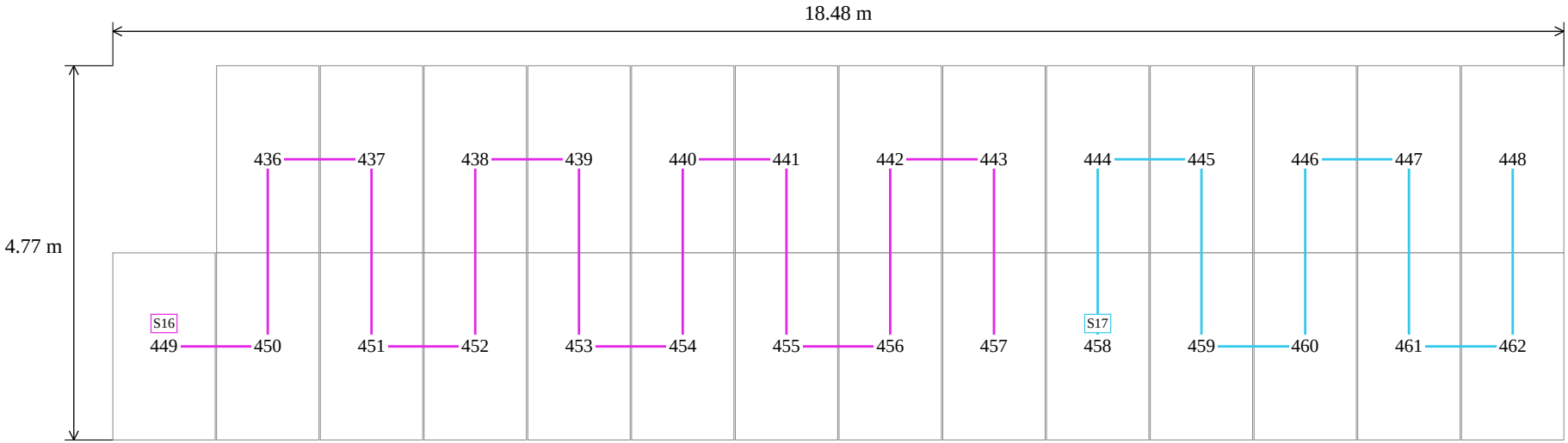
Zone 5 | 50 modules



Zone dimensions in metres; module IDs and string connections shown.

Physical block B6

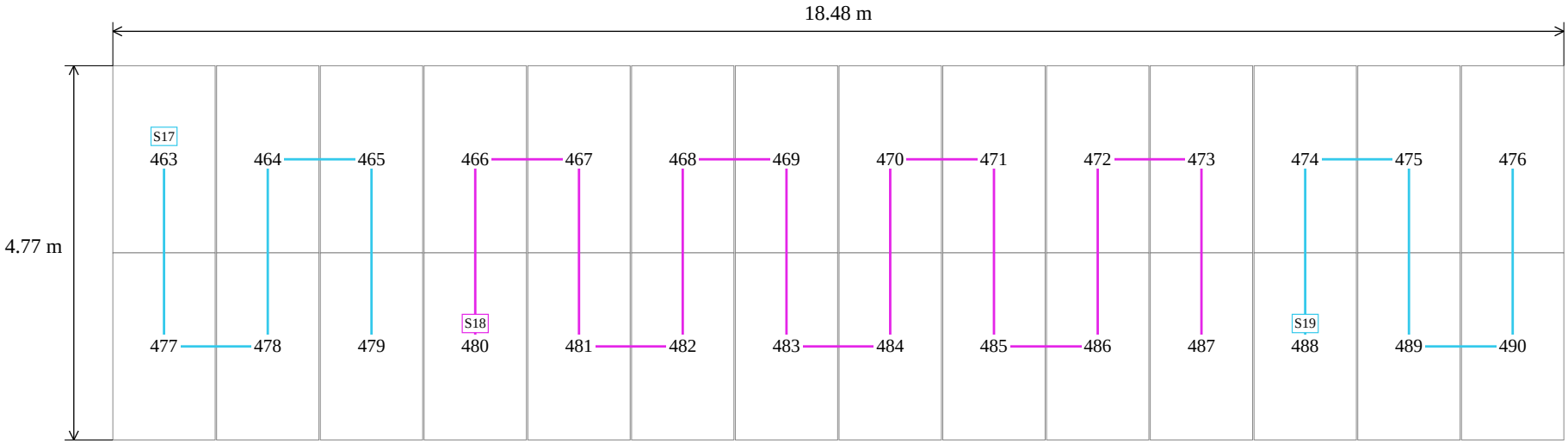
Zone 6 | 27 modules



Zone dimensions in metres; module IDs and string connections shown.

Physical block B7

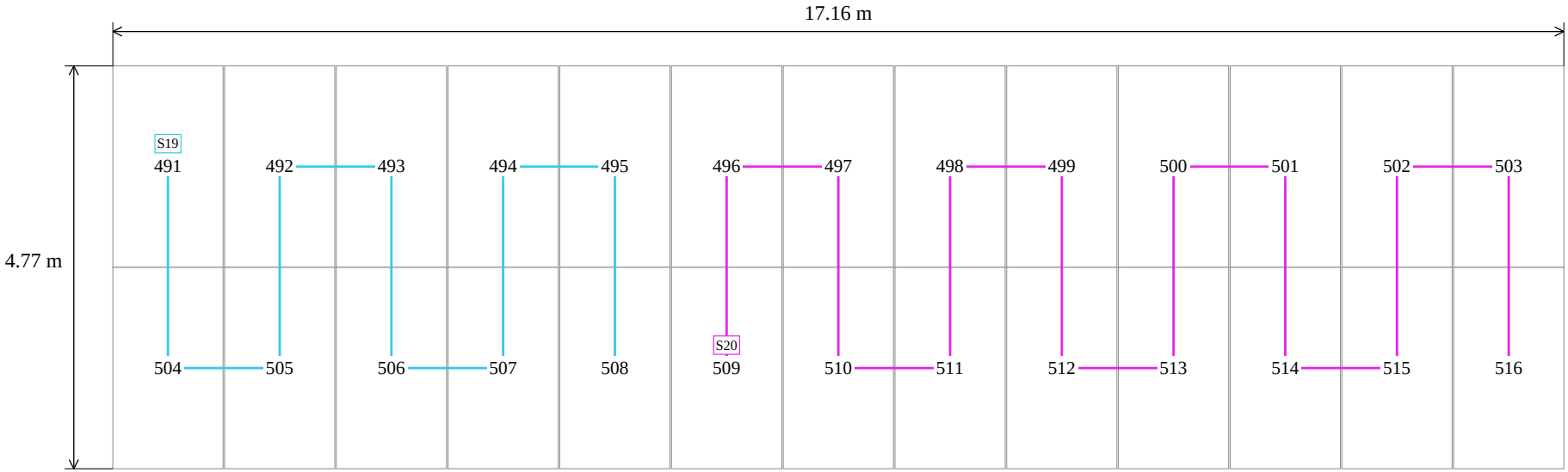
Zone 7 | 28 modules



Zone dimensions in metres; module IDs and string connections shown.

Physical block B8

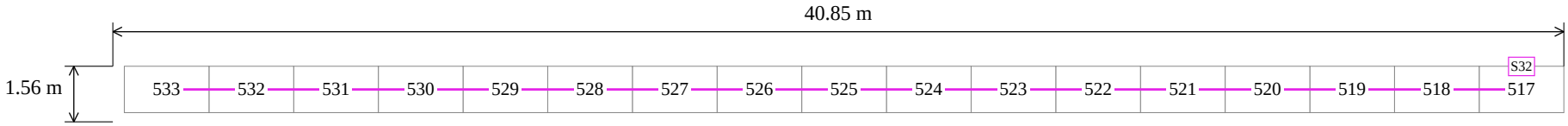
Zone 8 | 26 modules



Zone dimensions in metres; module IDs and string connections shown.

Physical block B9

Zone 9 | 17 modules



Zone dimensions in metres; module IDs and string connections shown.

Project notes and review items

Recorded notes and remaining engineering checks

Equipment data updated from supplied manufacturer documents: Trina Solar TSM-NEG21C.20 (720 W), Deye SUN-125K-SG02HP3-EU-GM10, Deye MC-L522 ESS brochure, and supplied MC-L522-BC + 125k H-INV system diagram. Module width corrected from 1333 mm to 1303 mm. Cached PV/BESS, shadow and 3D electrical-route results were cleared and must be recalculated. Important review item: existing String 32 contains 21 modules. With the corrected 720 W module data, its STC Vmp is about 867.3 V and Voc about 1037.4 V, above the inverter's 850 V MPPT upper limit and 1000 V maximum PV input voltage respectively. The string topology was not changed automatically.
2026-10-20 IMPLANTATION PROVISOIRE 533 modules : geometrie et cotes recalculees, zones repositionnees; certains retraits de rive < 1000mm (voir note Montage). Garder toutes les affectations et strings; cotes manuelles obsoletes retirees.

Review structural loads, fire access, string voltage, protection coordination, earthing and grid acceptance before construction.

Review item
No surveyed cable path: route lengths are preliminary estimates.
String 32: electrical validation incomplete for selected module / inverter.
String 32: hot Vmp check missing coefficient / maximum cell temperature.
String 1: electrical validation incomplete for selected module / inverter.
String 1: hot Vmp check missing coefficient / maximum cell temperature.
String 2: electrical validation incomplete for selected module / inverter.
String 2: hot Vmp check missing coefficient / maximum cell temperature.
String 3: electrical validation incomplete for selected module / inverter.
String 3: hot Vmp check missing coefficient / maximum cell temperature.
String 4: electrical validation incomplete for selected module / inverter.
String 4: hot Vmp check missing coefficient / maximum cell temperature.
String 5: electrical validation incomplete for selected module / inverter.
String 5: hot Vmp check missing coefficient / maximum cell temperature.
String 6: electrical validation incomplete for selected module / inverter.
String 6: hot Vmp check missing coefficient / maximum cell temperature.
String 7: electrical validation incomplete for selected module / inverter.
String 7: hot Vmp check missing coefficient / maximum cell temperature.
String 8: electrical validation incomplete for selected module / inverter.
String 8: hot Vmp check missing coefficient / maximum cell temperature.
String 9: electrical validation incomplete for selected module / inverter.

Review item
String 9: hot Vmp check missing coefficient / maximum cell temperature.
String 10: electrical validation incomplete for selected module / inverter.
String 10: hot Vmp check missing coefficient / maximum cell temperature.
String 11: electrical validation incomplete for selected module / inverter.
String 11: hot Vmp check missing coefficient / maximum cell temperature.
String 12: electrical validation incomplete for selected module / inverter.
String 12: hot Vmp check missing coefficient / maximum cell temperature.
String 15: electrical validation incomplete for selected module / inverter.
String 15: hot Vmp check missing coefficient / maximum cell temperature.
String 16: electrical validation incomplete for selected module / inverter.
String 16: hot Vmp check missing coefficient / maximum cell temperature.
String 17: electrical validation incomplete for selected module / inverter.
String 17: hot Vmp check missing coefficient / maximum cell temperature.
String 18: electrical validation incomplete for selected module / inverter.
String 18: hot Vmp check missing coefficient / maximum cell temperature.
String 19: electrical validation incomplete for selected module / inverter.
String 19: hot Vmp check missing coefficient / maximum cell temperature.
String 20: electrical validation incomplete for selected module / inverter.
String 20: hot Vmp check missing coefficient / maximum cell temperature.
String 21: electrical validation incomplete for selected module / inverter.
String 21: hot Vmp check missing coefficient / maximum cell temperature.
String 22: electrical validation incomplete for selected module / inverter.
String 22: hot Vmp check missing coefficient / maximum cell temperature.
String 23: electrical validation incomplete for selected module / inverter.
String 23: hot Vmp check missing coefficient / maximum cell temperature.
String 24: electrical validation incomplete for selected module / inverter.
String 24: hot Vmp check missing coefficient / maximum cell temperature.

Review item
String 25: electrical validation incomplete for selected module / inverter.
String 25: hot Vmp check missing coefficient / maximum cell temperature.
String 26: electrical validation incomplete for selected module / inverter.
String 26: hot Vmp check missing coefficient / maximum cell temperature.
String 27: electrical validation incomplete for selected module / inverter.
String 27: hot Vmp check missing coefficient / maximum cell temperature.
String 28: electrical validation incomplete for selected module / inverter.
String 28: hot Vmp check missing coefficient / maximum cell temperature.
String 29: electrical validation incomplete for selected module / inverter.
String 29: hot Vmp check missing coefficient / maximum cell temperature.
String 30: electrical validation incomplete for selected module / inverter.
String 30: hot Vmp check missing coefficient / maximum cell temperature.
String 31: electrical validation incomplete for selected module / inverter.
String 31: hot Vmp check missing coefficient / maximum cell temperature.
String 13: electrical validation incomplete for selected module / inverter.
String 13: hot Vmp check missing coefficient / maximum cell temperature.
String 14: electrical validation incomplete for selected module / inverter.
String 14: hot Vmp check missing coefficient / maximum cell temperature.
INV2 MPPT 9: current check incomplete.
INV2 MPPT 9: current check incomplete.
INV2 MPPT 9: per-input short-circuit limit not supplied.
INV1 MPPT 1: current check incomplete.
INV1 MPPT 1: current check incomplete.
INV1 MPPT 1: per-input short-circuit limit not supplied.
INV1 MPPT 2: current check incomplete.
INV1 MPPT 2: current check incomplete.
INV1 MPPT 2: per-input short-circuit limit not supplied.

Review item
INV1 MPPT 3: current check incomplete.
INV1 MPPT 3: current check incomplete.
INV1 MPPT 3: per-input short-circuit limit not supplied.
INV1 MPPT 4: current check incomplete.
INV1 MPPT 4: current check incomplete.
INV1 MPPT 4: per-input short-circuit limit not supplied.
INV1 MPPT 5: current check incomplete.
INV1 MPPT 5: current check incomplete.
INV1 MPPT 5: per-input short-circuit limit not supplied.
INV1 MPPT 6: current check incomplete.
INV1 MPPT 6: current check incomplete.
INV1 MPPT 6: per-input short-circuit limit not supplied.
INV1 MPPT 7: current check incomplete.
INV1 MPPT 7: current check incomplete.
INV1 MPPT 7: per-input short-circuit limit not supplied.
INV1 MPPT 8: current check incomplete.
INV1 MPPT 8: current check incomplete.
INV1 MPPT 8: per-input short-circuit limit not supplied.
INV1 MPPT 9: current check incomplete.
INV1 MPPT 9: current check incomplete.
INV1 MPPT 9: per-input short-circuit limit not supplied.
INV1 MPPT 10: current check incomplete.
INV1 MPPT 10: current check incomplete.
INV1 MPPT 10: per-input short-circuit limit not supplied.
INV2 MPPT 3: current check incomplete.
INV2 MPPT 3: current check incomplete.
INV2 MPPT 3: per-input short-circuit limit not supplied.

Review item
INV2 MPPT 4: current check incomplete.
INV2 MPPT 4: current check incomplete.
INV2 MPPT 4: per-input short-circuit limit not supplied.
INV2 MPPT 5: current check incomplete.
INV2 MPPT 5: current check incomplete.
INV2 MPPT 5: per-input short-circuit limit not supplied.
INV2 MPPT 6: current check incomplete.
INV2 MPPT 6: current check incomplete.
INV2 MPPT 6: per-input short-circuit limit not supplied.
INV2 MPPT 7: current check incomplete.
INV2 MPPT 7: current check incomplete.
INV2 MPPT 7: per-input short-circuit limit not supplied.
INV2 MPPT 8: current check incomplete.
INV2 MPPT 8: current check incomplete.
INV2 MPPT 8: per-input short-circuit limit not supplied.
INV3 MPPT 1: current check incomplete.
INV3 MPPT 1: current check incomplete.
INV3 MPPT 1: per-input short-circuit limit not supplied.
INV3 MPPT 2: current check incomplete.
INV3 MPPT 2: current check incomplete.
INV3 MPPT 2: per-input short-circuit limit not supplied.
INV3 MPPT 3: current check incomplete.
INV3 MPPT 3: current check incomplete.
INV3 MPPT 3: per-input short-circuit limit not supplied.
INV3 MPPT 4: current check incomplete.
INV3 MPPT 4: current check incomplete.
INV3 MPPT 4: per-input short-circuit limit not supplied.

Review item
INV3 MPPT 5: current check incomplete.
INV3 MPPT 5: current check incomplete.
INV3 MPPT 5: per-input short-circuit limit not supplied.
INV3 MPPT 6: current check incomplete.
INV3 MPPT 6: current check incomplete.
INV3 MPPT 6: per-input short-circuit limit not supplied.
INV3 MPPT 7: current check incomplete.
INV3 MPPT 7: current check incomplete.
INV3 MPPT 7: per-input short-circuit limit not supplied.
INV3 MPPT 8: current check incomplete.
INV3 MPPT 8: current check incomplete.
INV3 MPPT 8: per-input short-circuit limit not supplied.
INV3 MPPT 9: current check incomplete.
INV3 MPPT 9: current check incomplete.
INV3 MPPT 9: per-input short-circuit limit not supplied.
INV3 MPPT 10: current check incomplete.
INV3 MPPT 10: current check incomplete.
INV3 MPPT 10: per-input short-circuit limit not supplied.
INV2 MPPT 1: current check incomplete.
INV2 MPPT 1: current check incomplete.
INV2 MPPT 1: per-input short-circuit limit not supplied.
INV2 MPPT 2: current check incomplete.
INV2 MPPT 2: current check incomplete.
INV2 MPPT 2: per-input short-circuit limit not supplied.
INV2: maximum PV input power not supplied.
INV1: maximum PV input power not supplied.
INV3: maximum PV input power not supplied.

Review item
Final checks required: hot Vmp, bifacial current gain, per-input limits, ampacity, AC/BESS protections and grid certification.

Cable inventory and mass

DC string routes plus declared additional cables

Cable	Length m	kg/m	Quantity

Unlisted cables are outside the inventory. Missing lengths or manufacturer masses are unavailable, not zero.

Mass input	Value
total_kg	
known_kg	0
missing	32

Economic inputs

Recorded assumptions

Parameter	Value
import_eur_kwh	0.25
export_low_eur_kwh	0.07
export_high_eur_kwh	0.08
pv_capex_eur	250,000
bess_capex_each_eur	50,000
annual_maintenance_eur	0
discount_rate_pct	0
annual_degradation_pct	0
lifetime_years	15
replacement_year	0
replacement_cost_eur	0